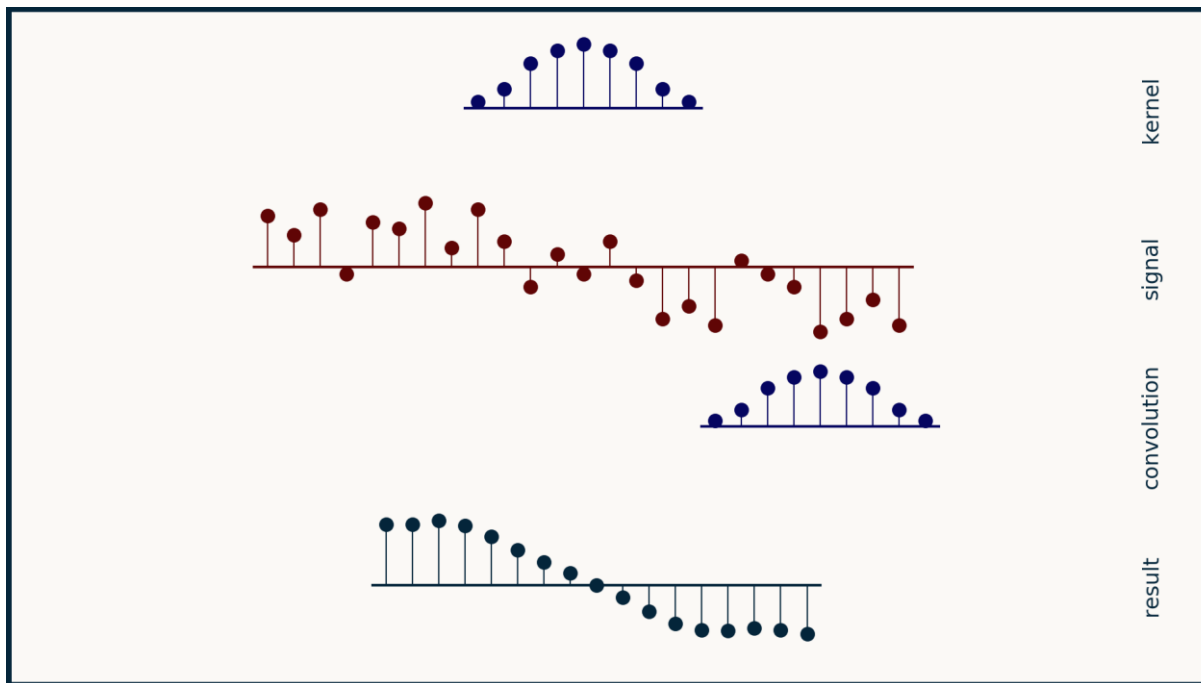


Signals and Systems

Lecture 6 : Convolution

Lecture 7 : Correlation



Convolution

■ Definition of convolution

$$y[n] = x[n] * h[n] = \sum_{m=-\infty}^{\infty} X[m] h[n - m]$$

↑ ↑
Shift that Domain
corresponds that we
to the final multiply
point and sum
 over

Signal Operations

■ Consider this

- Plotting $x[3 - n]$
- We follow this procedure

$$\diamond x[n] \rightarrow x[n + 3] \rightarrow x[-n + 3] = x[3 - n]$$

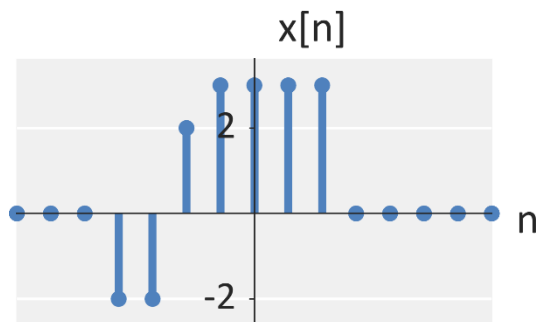
$\uparrow$ $\uparrow$
Shift Time
left 3 reverse

Signal Operations

■ Consider this

- Plotting $x[3 - n]$
- We follow this procedure

$$\diamond x[n] \xrightarrow{\substack{\uparrow \\ \text{Shift} \\ \text{left } 3}} x[n + 3] \xrightarrow{\substack{\uparrow \\ \text{Time} \\ \text{reverse}}} x[-n + 3] = x[3 - n]$$

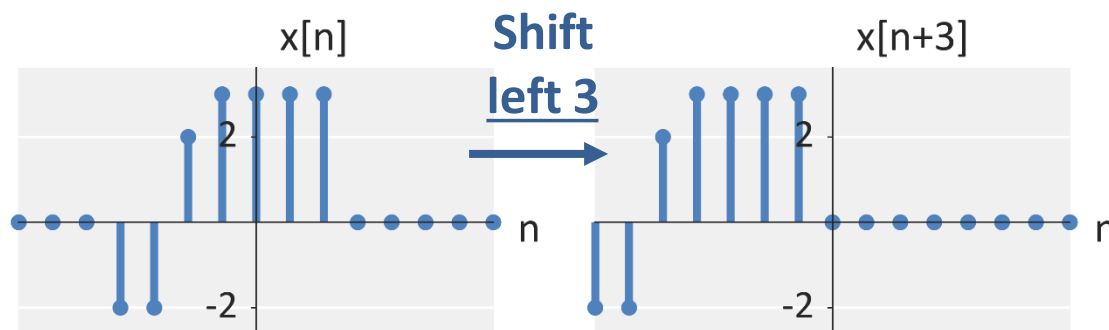


Signal Operations

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- Plotting $x[3 - n]$
- We follow this procedure

$$\diamond x[n] \xrightarrow{\substack{\uparrow \\ \text{Shift} \\ \text{left } 3}} x[n+3] \xrightarrow{\substack{\uparrow \\ \text{Time} \\ \text{reverse}}} x[-n+3] = x[3-n]$$

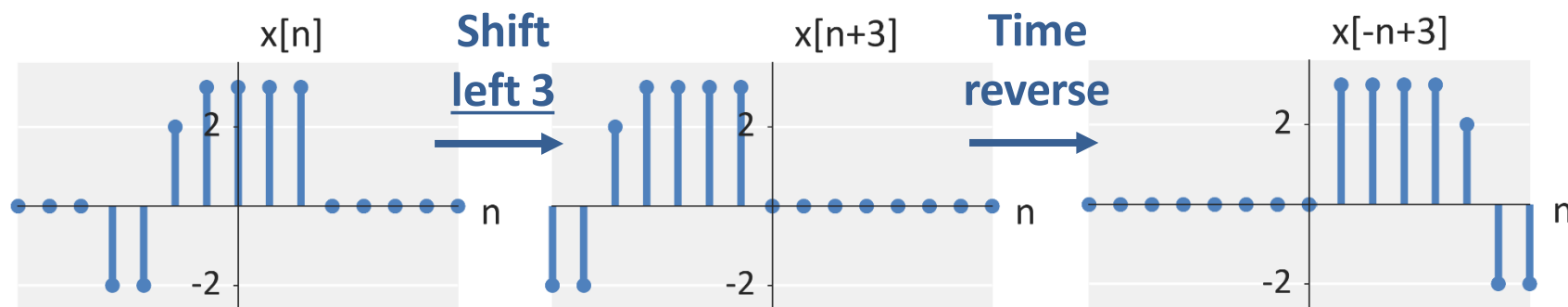


Signal Operations

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- Plotting $x[3 - n]$
- We follow this procedure

$$\diamond x[n] \xrightarrow{\substack{\uparrow \\ \text{Shift} \\ \text{left } 3}} x[n+3] \xrightarrow{\substack{\uparrow \\ \text{Time} \\ \text{reverse}}} x[-n+3] = x[3-n]$$



Signal Operations

■ Consider this

- Plotting $x[3 - n]$
- We follow this procedure

$$\diamond x[n] \xrightarrow[\text{Shift left 3}]{\uparrow} x[n + 3] \xrightarrow[\text{Time reverse}]{\uparrow} x[-n + 3] = x[3 - n]$$

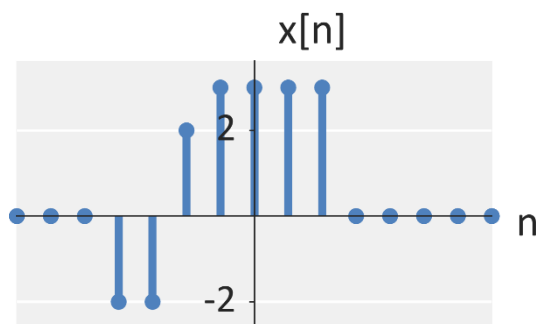
- We can go in the other direction (used to illustrate convolution)

$$\diamond x[n] \xrightarrow[\text{Time reverse}]{\uparrow} x[-n] \xrightarrow[\text{Shift right 3}]{\uparrow} x[-(n - 3)] = x[-n + 3] = x[3 - n]$$

Signal Operations

■ Consider this

- Plotting $x[3 - n]$



- We can go in the other direction (used to illustrate convolution)

$$\diamond x[n] \rightarrow x[-n] \rightarrow x[-(n - 3)] = x[-n + 3] = x[3 - n]$$

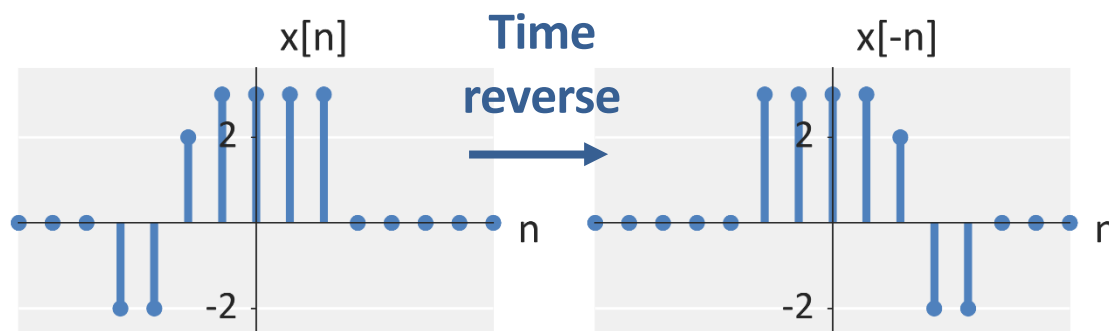
↑
Time
reverse

↑
Shift
right 3

Signal Operations

■ Consider this

- Plotting $x[3 - n]$



- We can go in the other direction (used to illustrate convolution)

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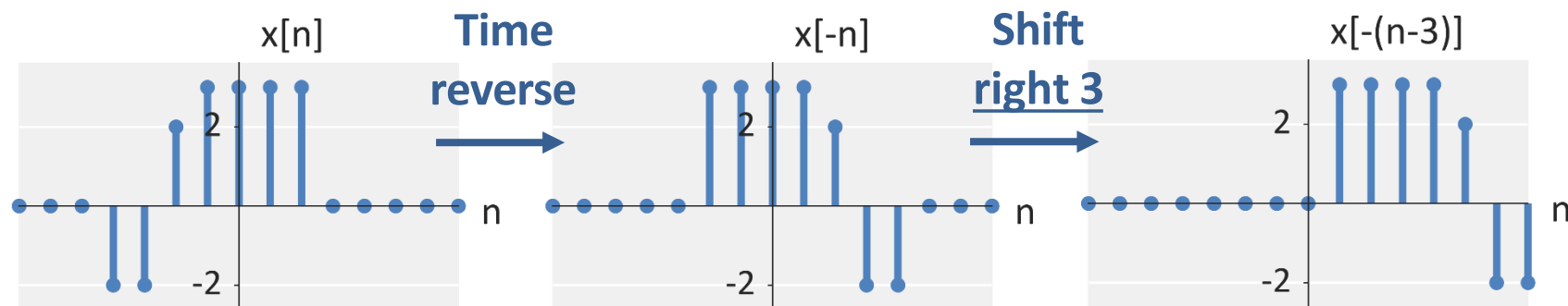
Time
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Signal Operations

■ Consider this

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$$\diamond x[n] \rightarrow x[-n] \rightarrow x[-(n-3)] = x[-n+3] = x[3-n]$$

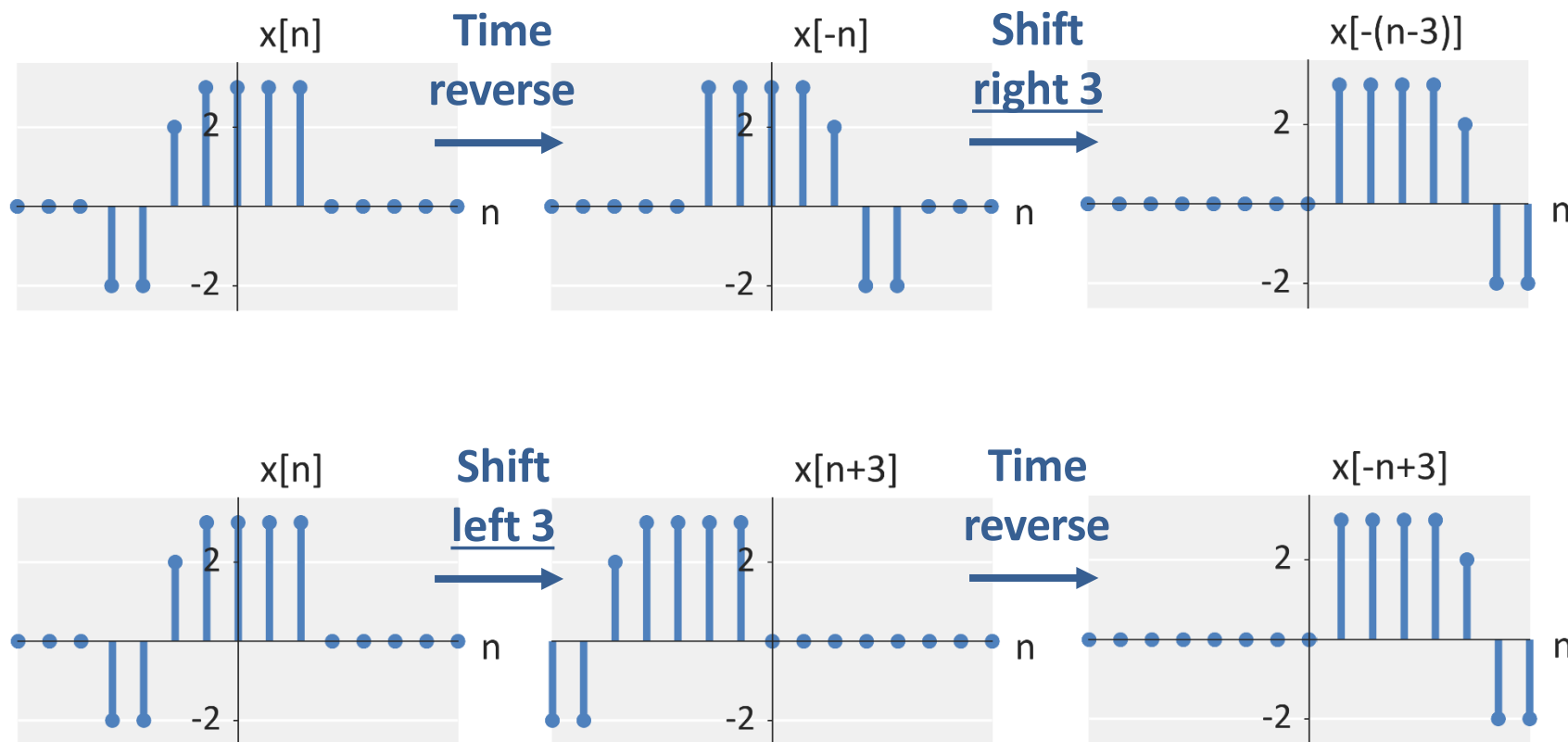
↑
Time
reverse

↑
Shift
right 3

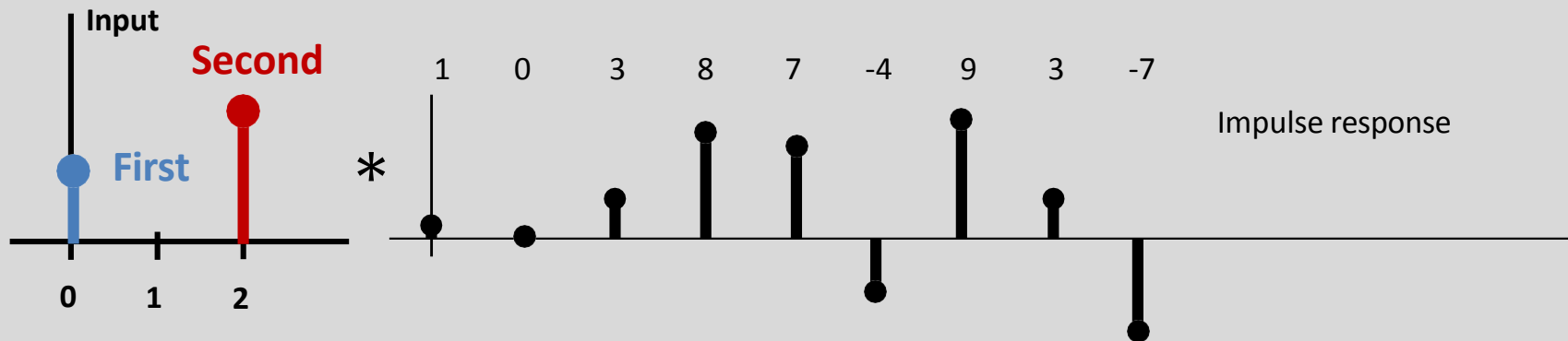
Signal Operations

■ Consider this

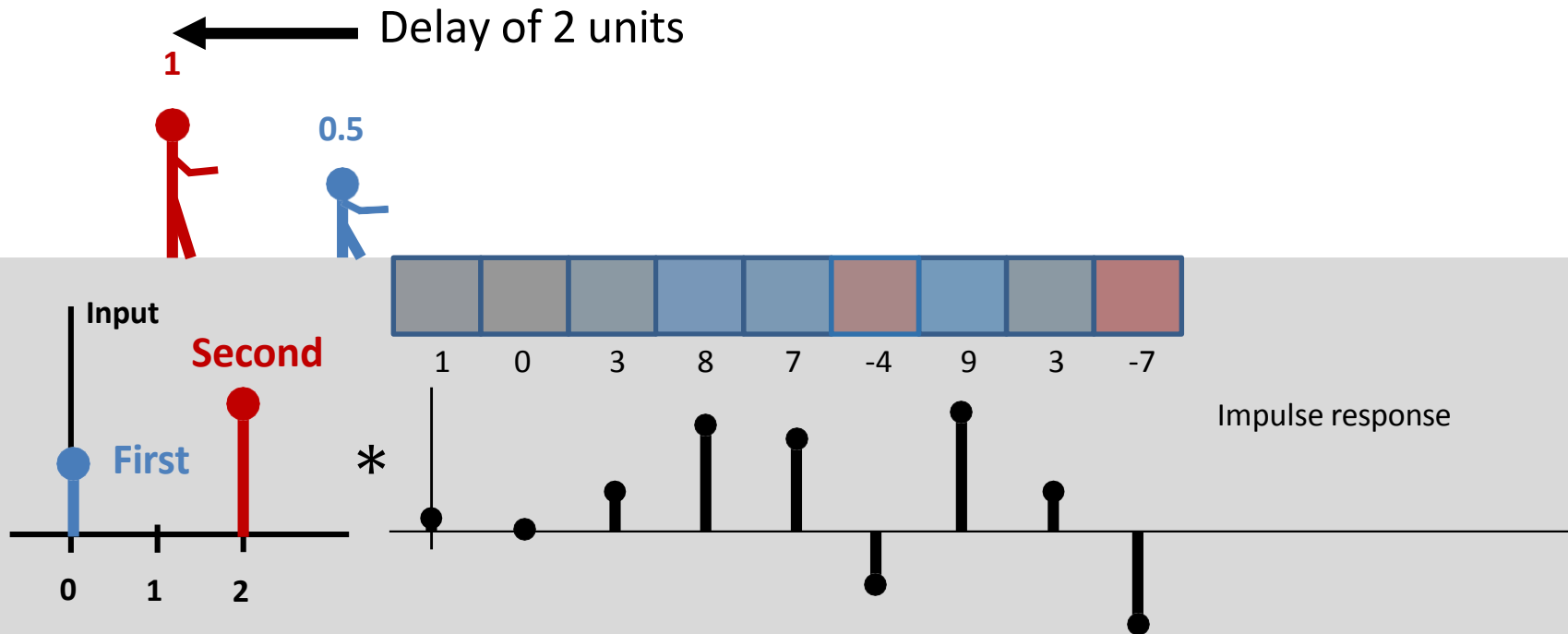
- Plotting $x[3 - n]$



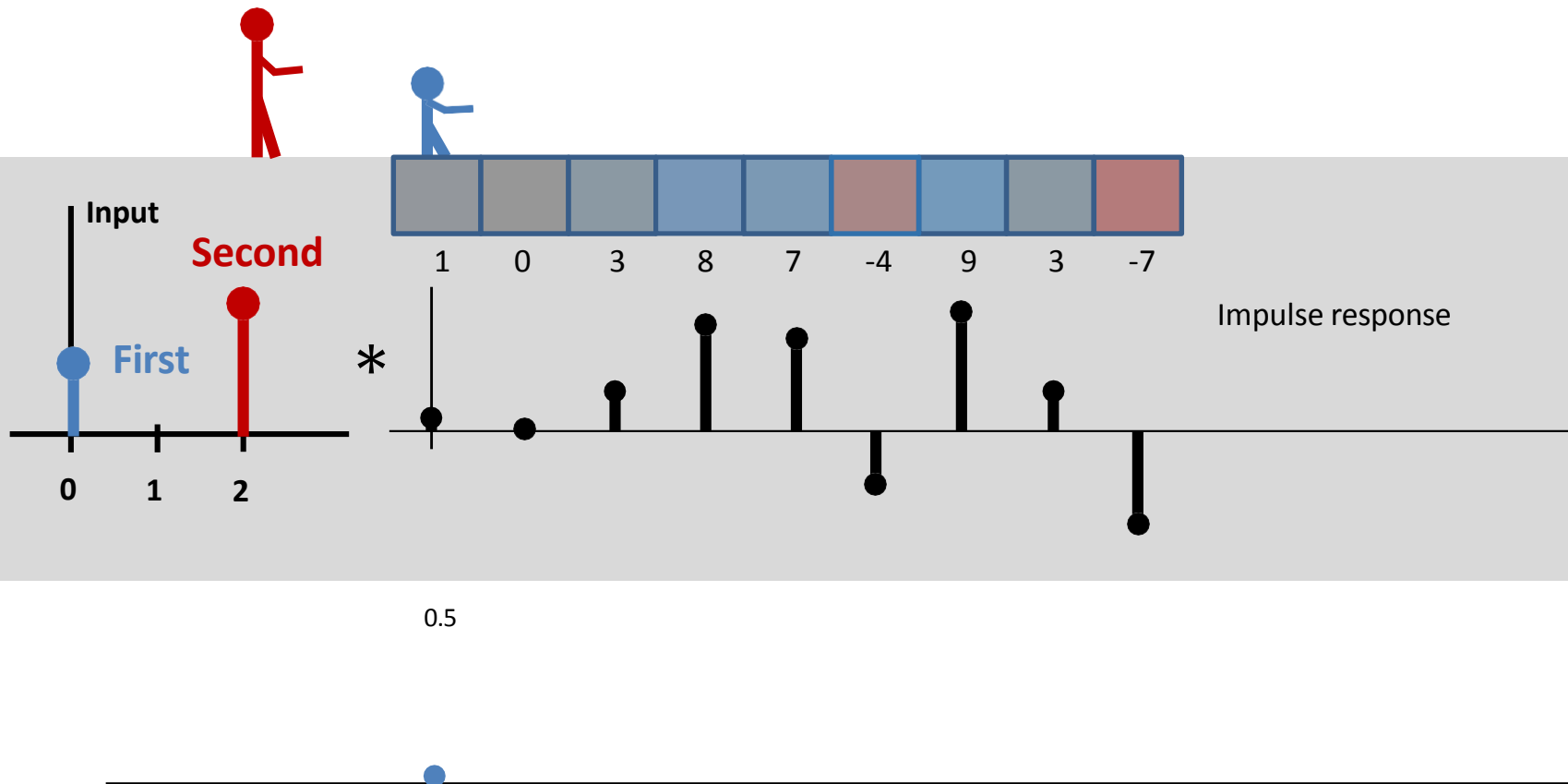
■ **Convolution:** $x[n] * h[n] = \sum_{m=-\infty}^{\infty} x[m] h[n - m]$



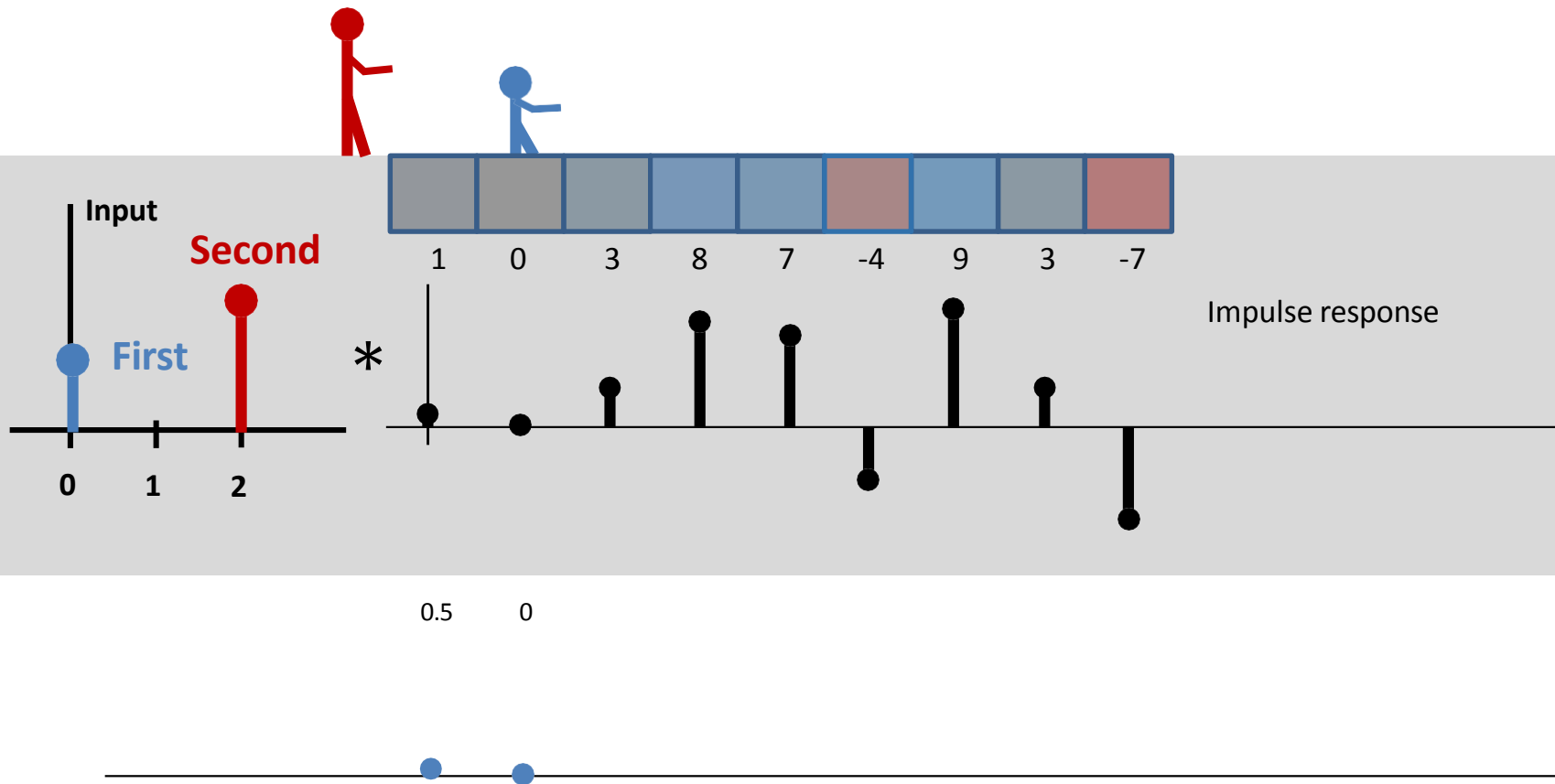
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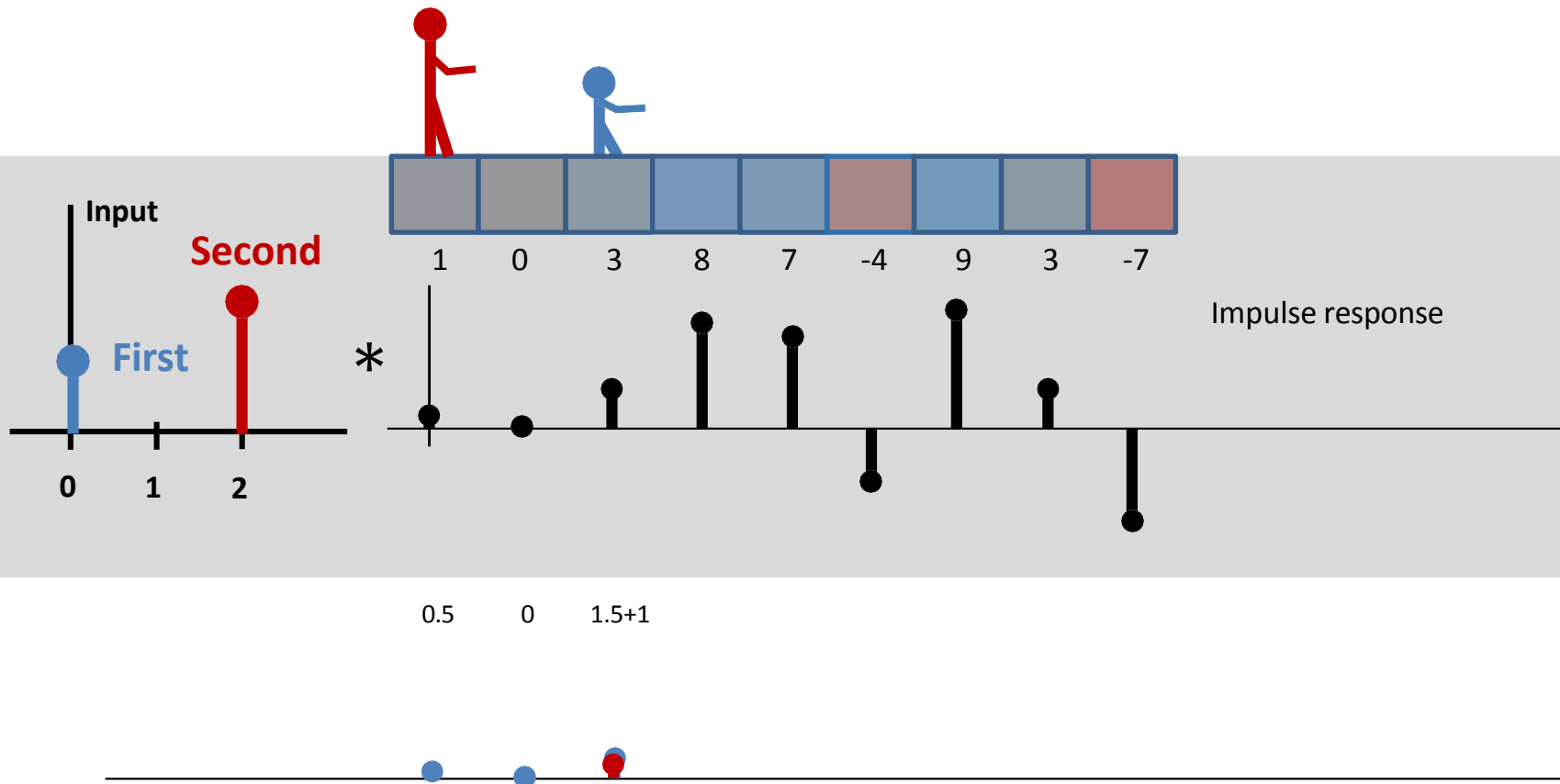
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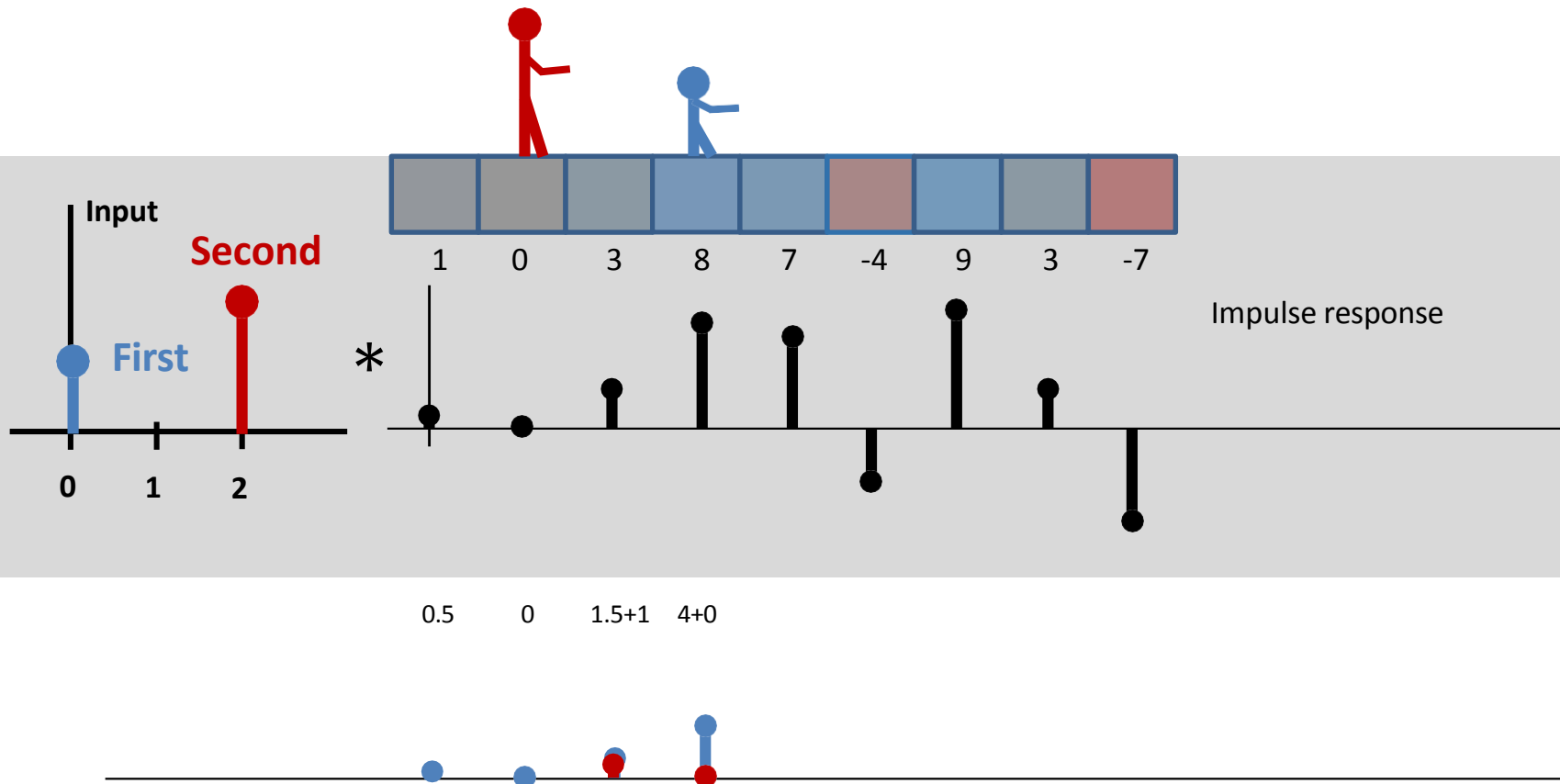
■ **Convolution:** $x[n] * h[n] = \sum_{m=-\infty}^{\infty} x[m] h[n - m]$



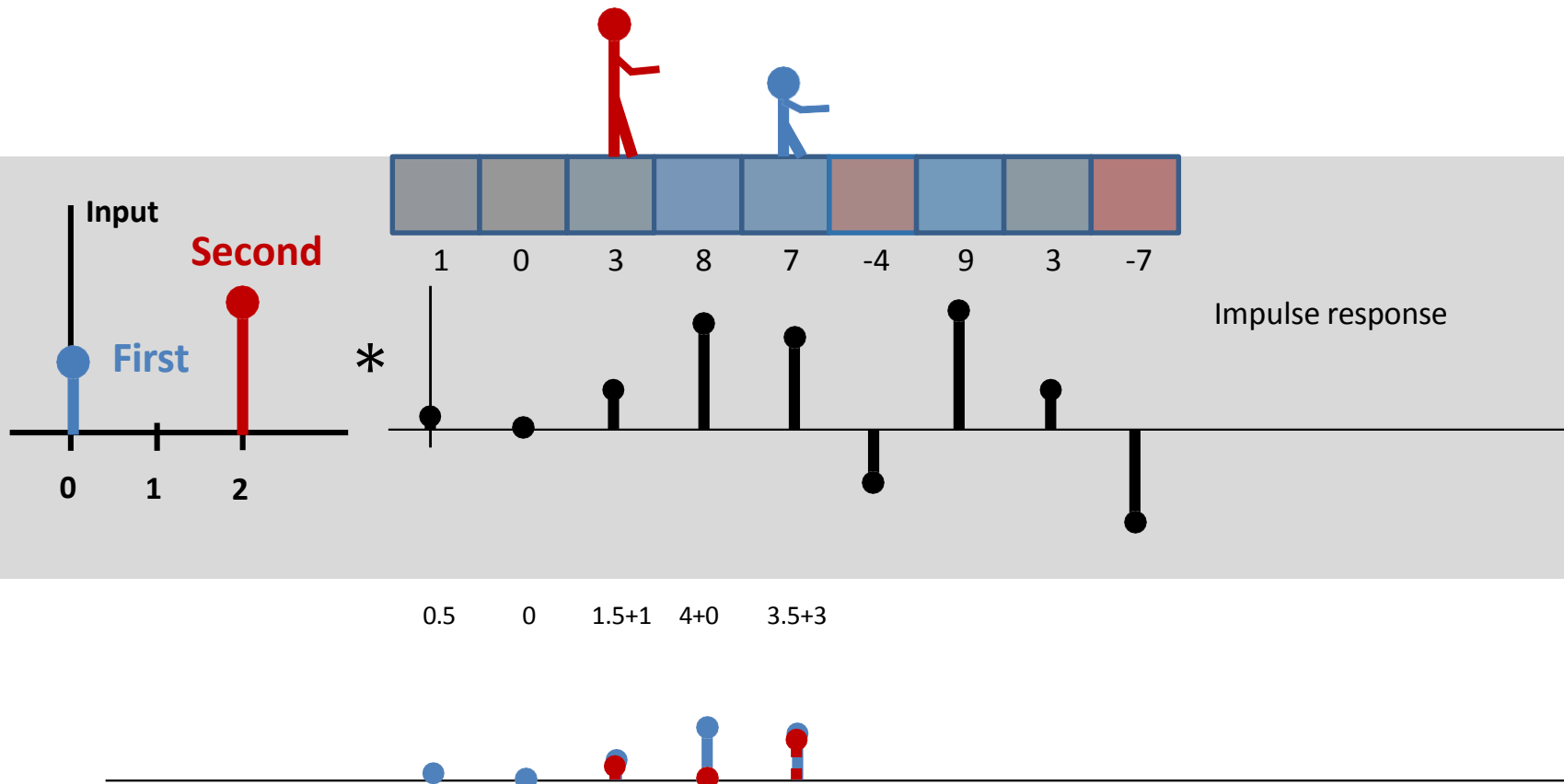
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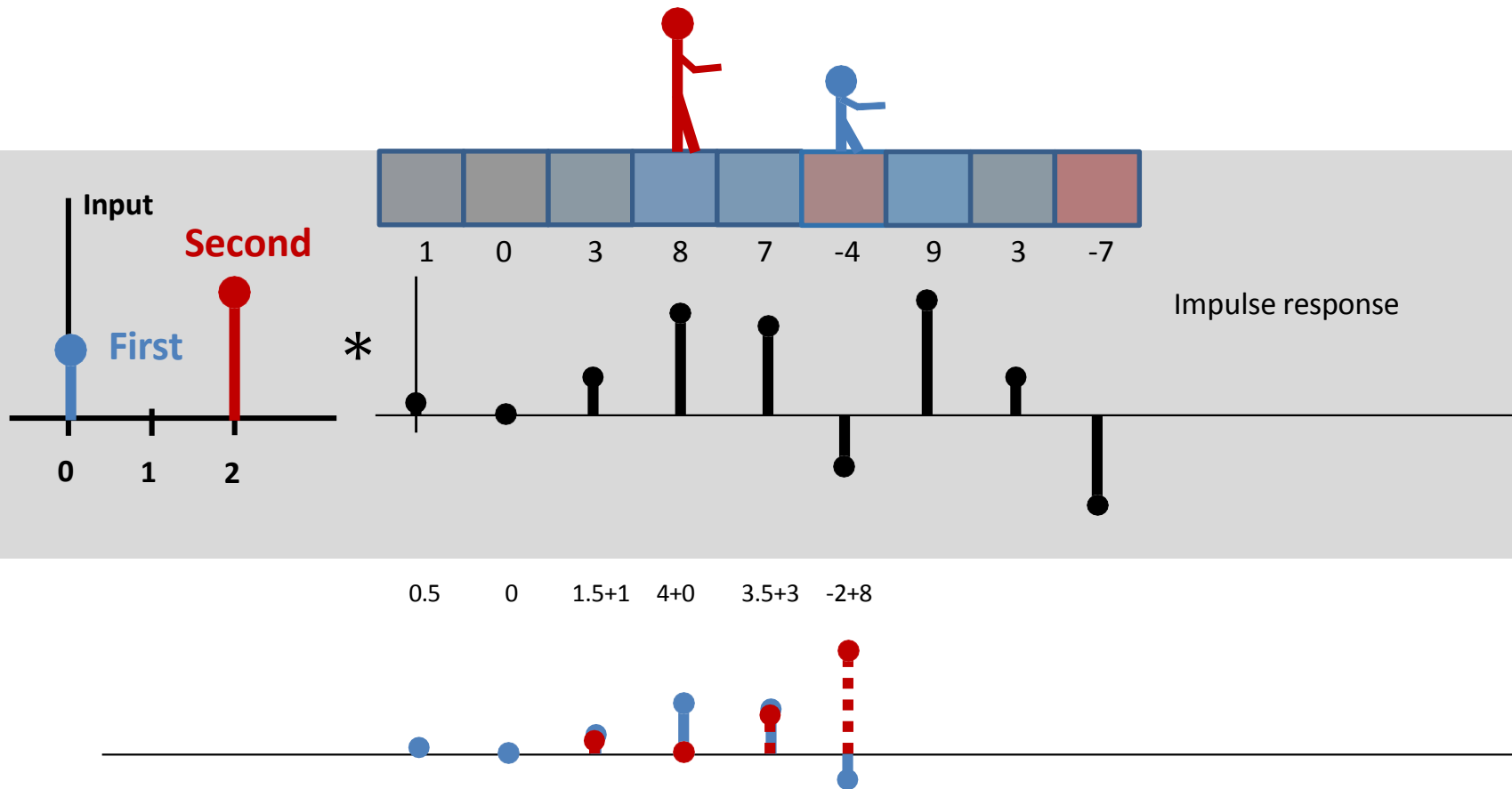
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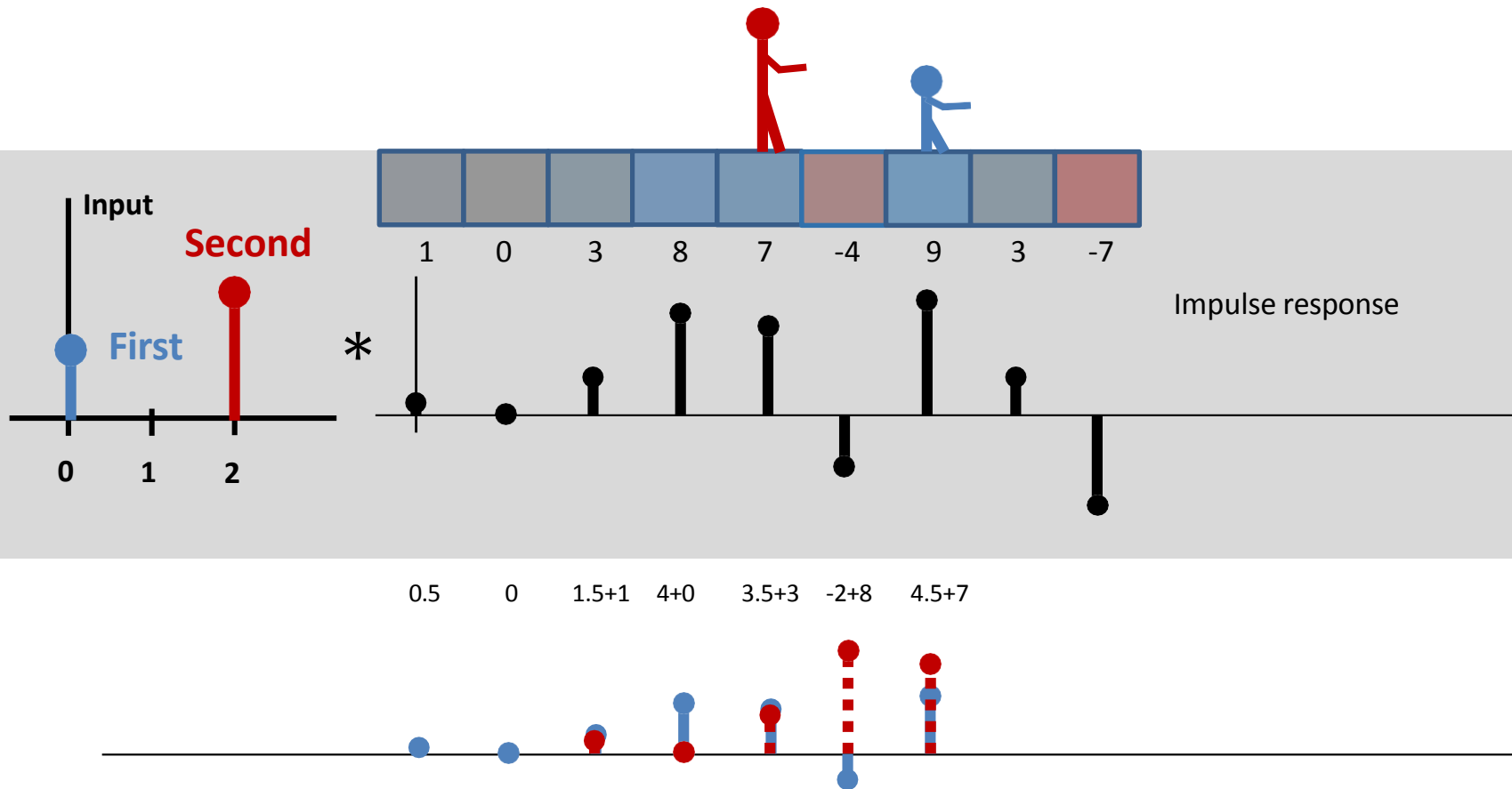
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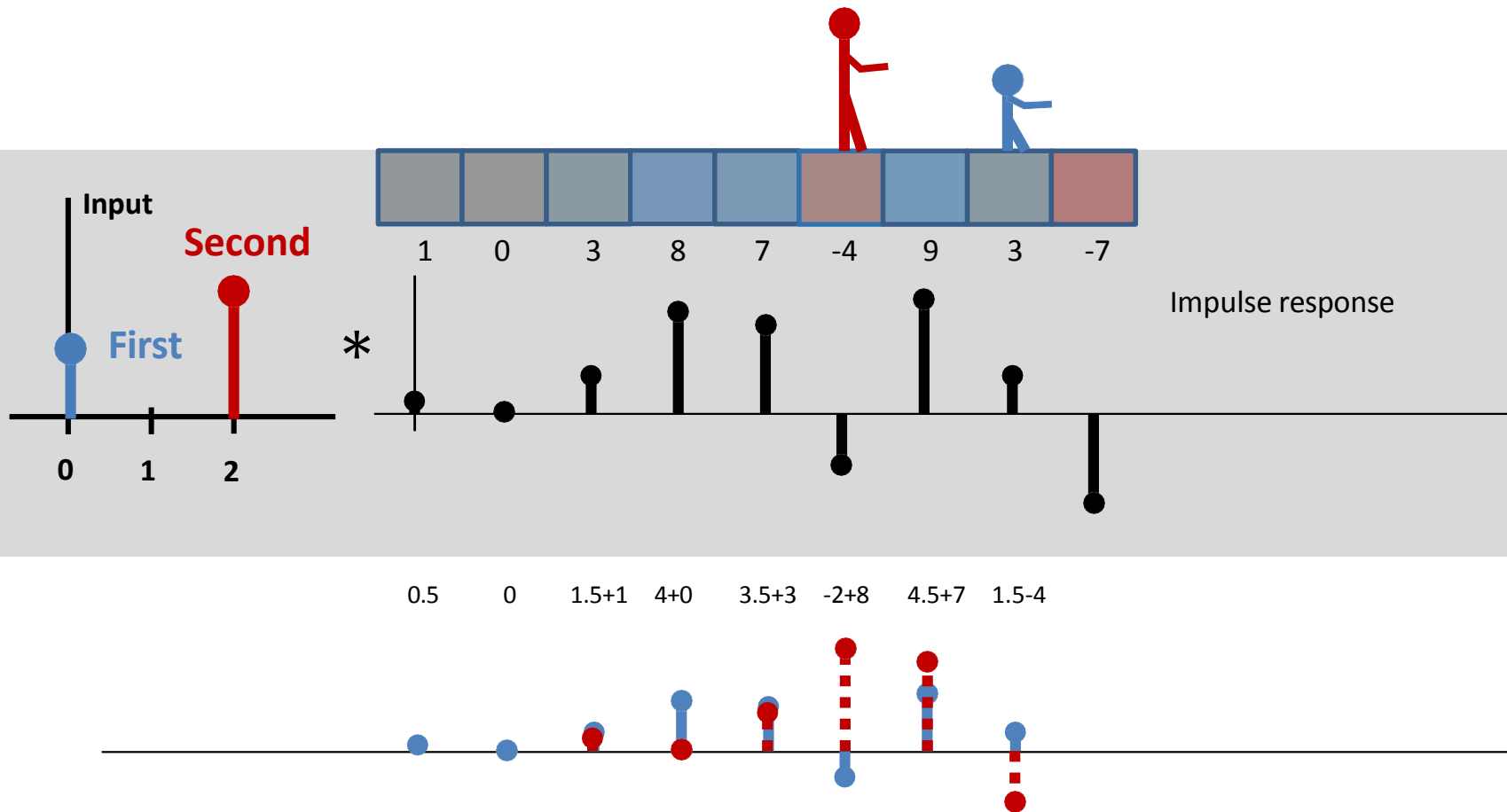
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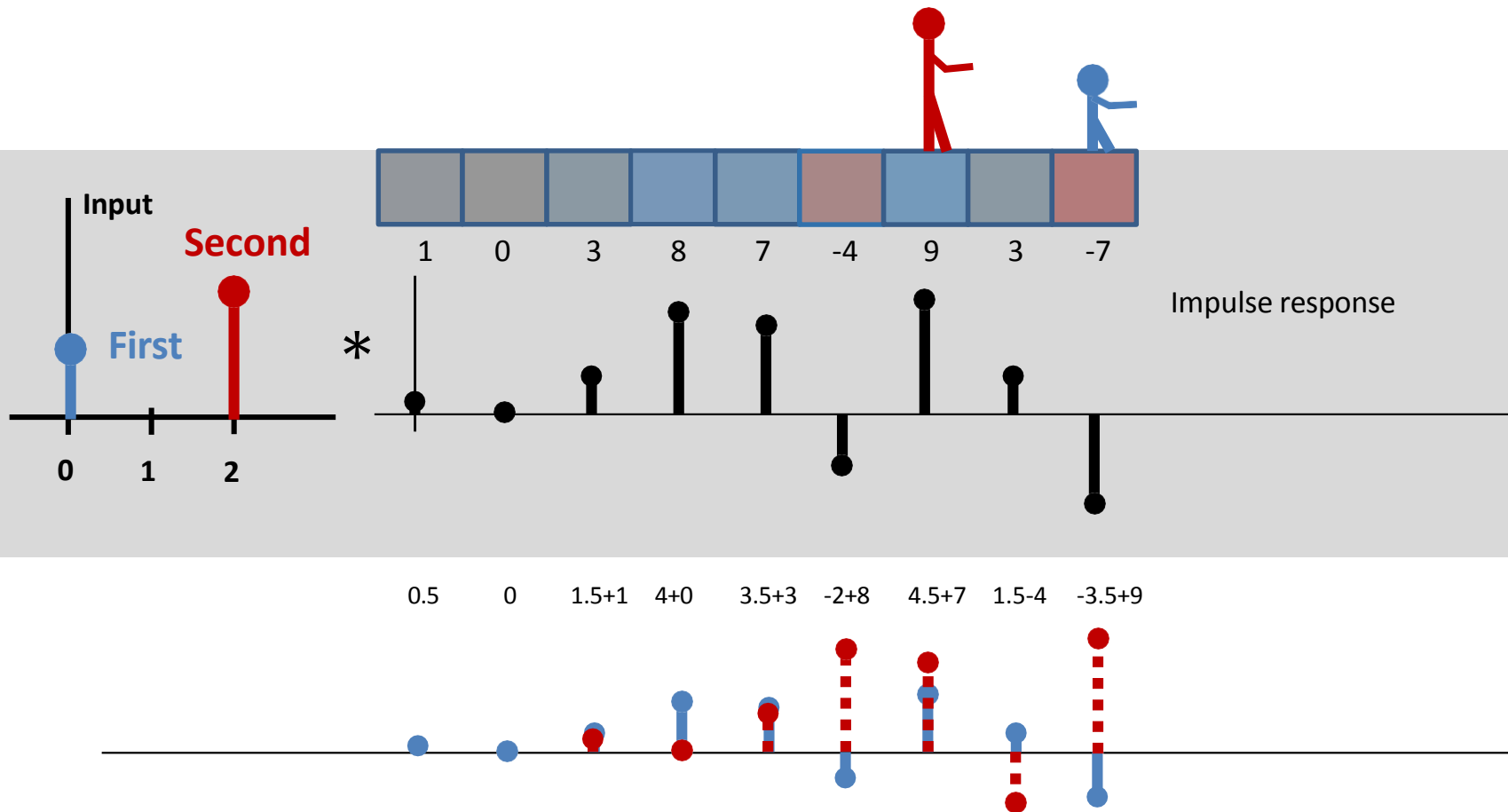
■ **Convolution:** $x[n] * h[n] = \sum_{m=-\infty}^{\infty} x[m] h[n - m]$



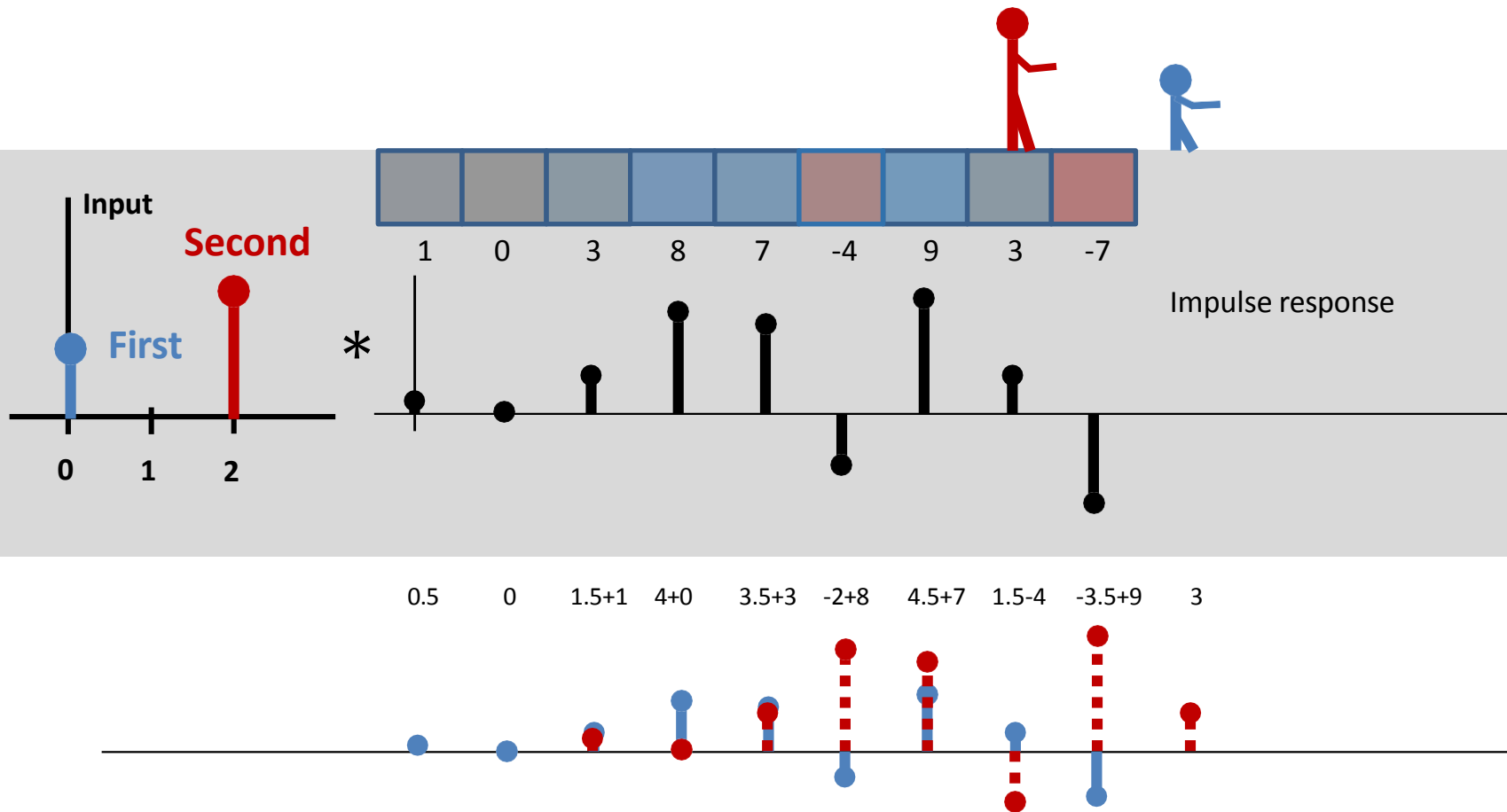
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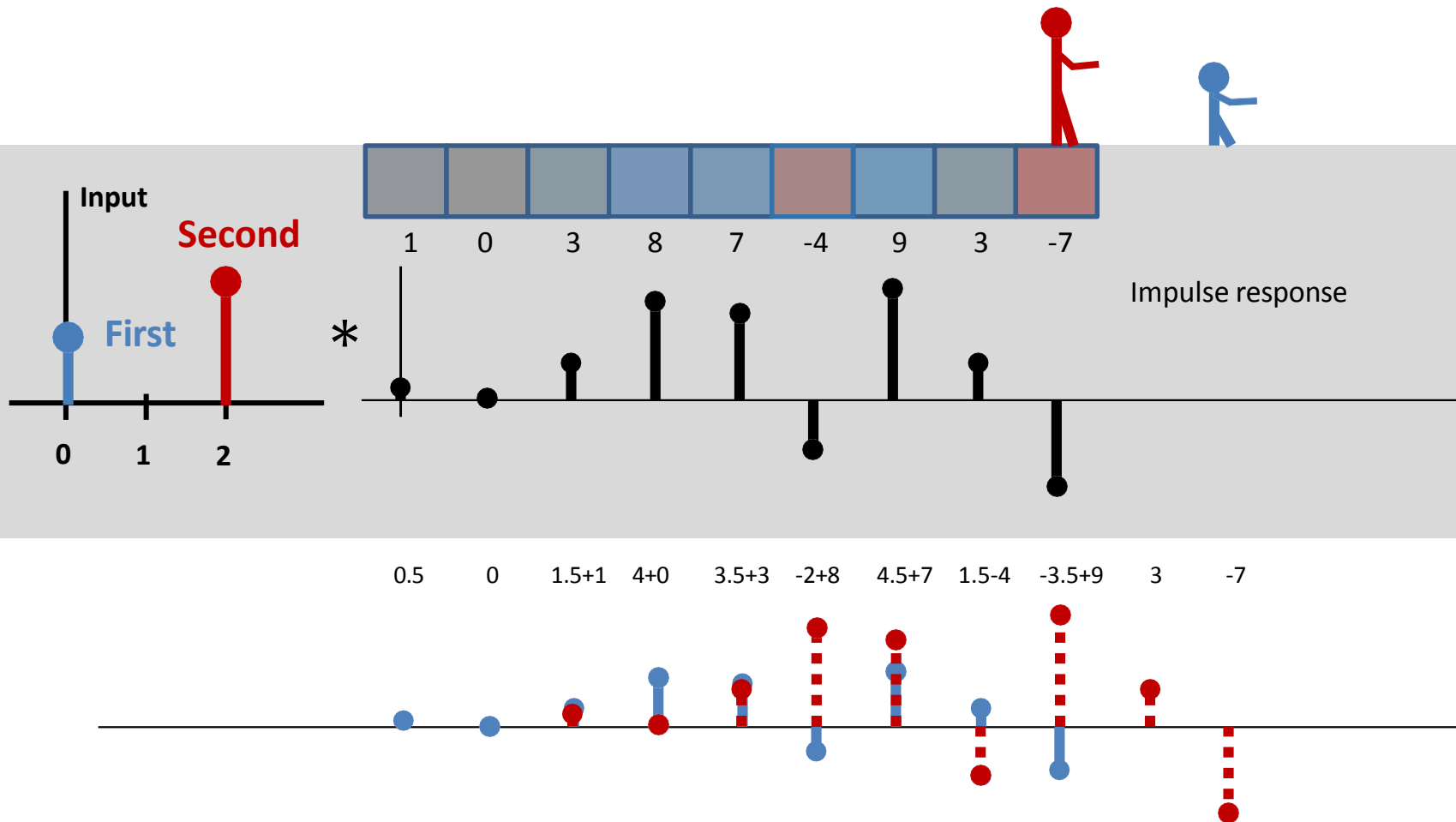
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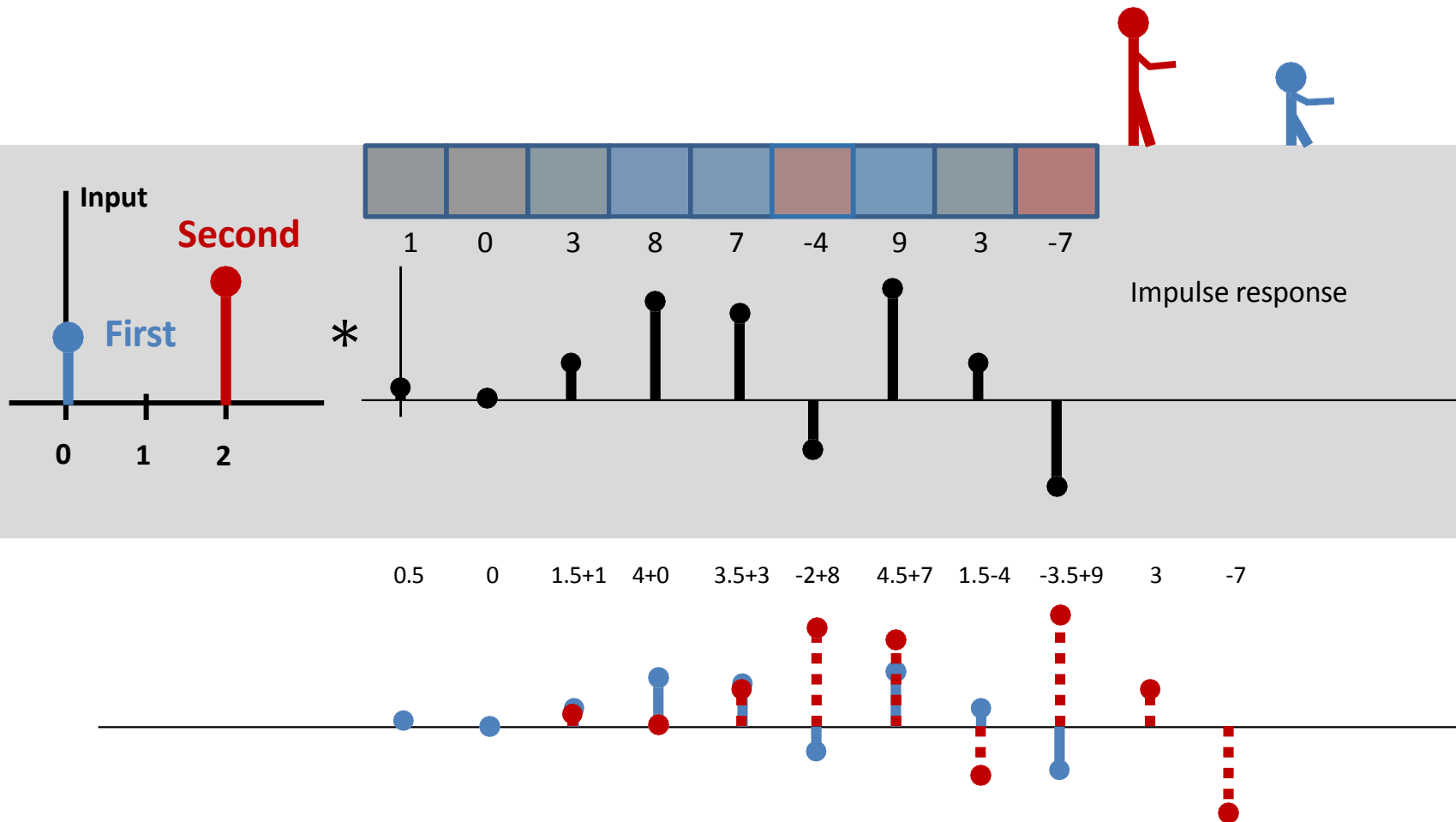
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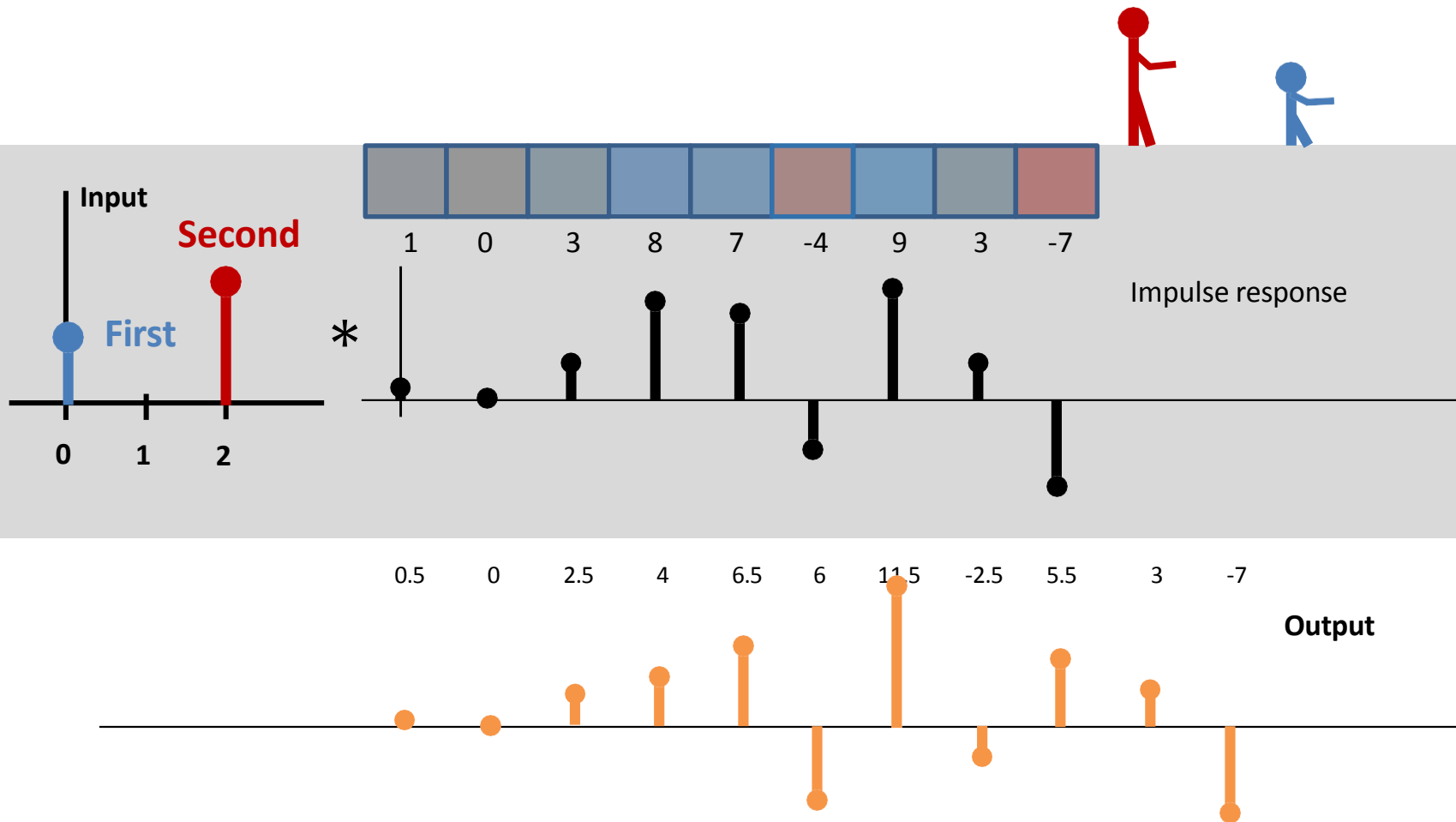
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■ **Convolution:** $x[n] * h[n] = \sum_{m=-\infty}^{\infty} x[m] h[n - m]$



■ Linear and Time-Invariant (LTI) System

- Consider the system with input-output relationship:

$$y[n] = \frac{1}{2} (x[n] + x[n - 1])$$

- Compute response of the system to input of

$$x[n] = 2 \delta[n - 1] + 2 \delta[n - 2]$$

Convolution

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- **Solution:**

$$\begin{aligned} y[n] &= \frac{1}{2} (2 \delta[n-1] + 2 \delta[n-2] + 2 \delta[n-2] + 2 \delta[n-3]) \\ &= \frac{1}{2} (2 \delta[n-1] + 4 \delta[n-2] + 2 \delta[n-3]) \\ &= \delta[n-1] + 2 \delta[n-2] + \delta[n-3] \end{aligned}$$

Convolution

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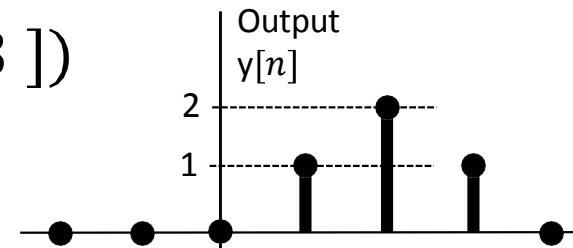
$$x[n] = 2 \delta[n-1] + 2 \delta[n-2]$$

- **Solution:**

$$y[n] = \frac{1}{2} (2 \delta[n-1] + 2 \delta[n-2] + 2 \delta[n-2] + 2 \delta[n-3])$$

$$= \frac{1}{2} (2 \delta[n-1] + 4 \delta[n-2] + 2 \delta[n-3])$$

$$= \delta[n-1] + 2 \delta[n-2] + \delta[n-3]$$



Convolution

■ Linear and Time-Invariant (LTI) System

- Consider the system with input-output relationship:

$$y[n] = \frac{1}{2} (x[n] + x[n - 1])$$

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Convolution

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$$y[n] = \frac{1}{2} (x[n] + x[n-1])$$

- Compute the impulse response $h[n]$ of the system

- **Solution:**

$$x[n] = \delta[n]$$

$$h[n] = \frac{1}{2} (\delta[n] + \delta[n-1])$$

Convolution

■ Linear and Time-Invariant (LTI) System

- Consider the system with input-output relationship:

$$y[n] = \frac{1}{2} (x[n] + x[n - 1])$$

- Compute the convolution of:

$$y[n] = h[n] * x[n]$$

$$h[n] = (1/2)(\delta[n] + \delta[n - 1])$$

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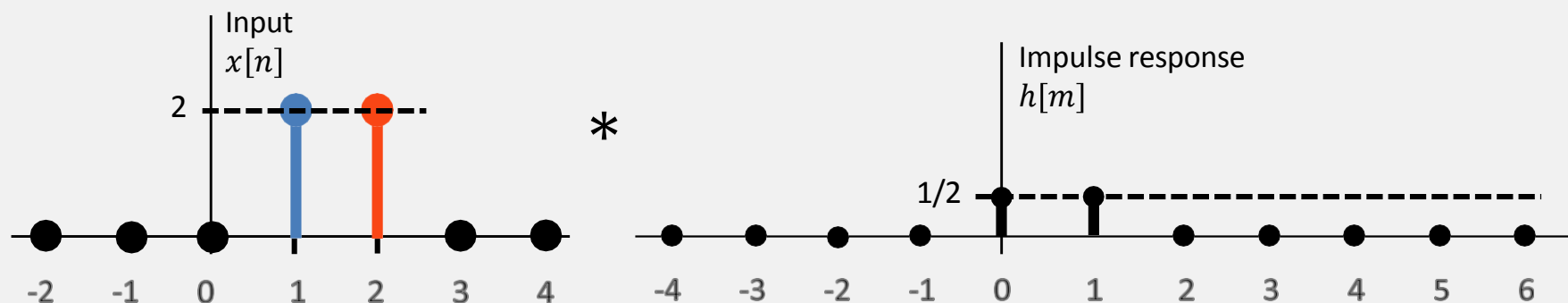
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Convolution

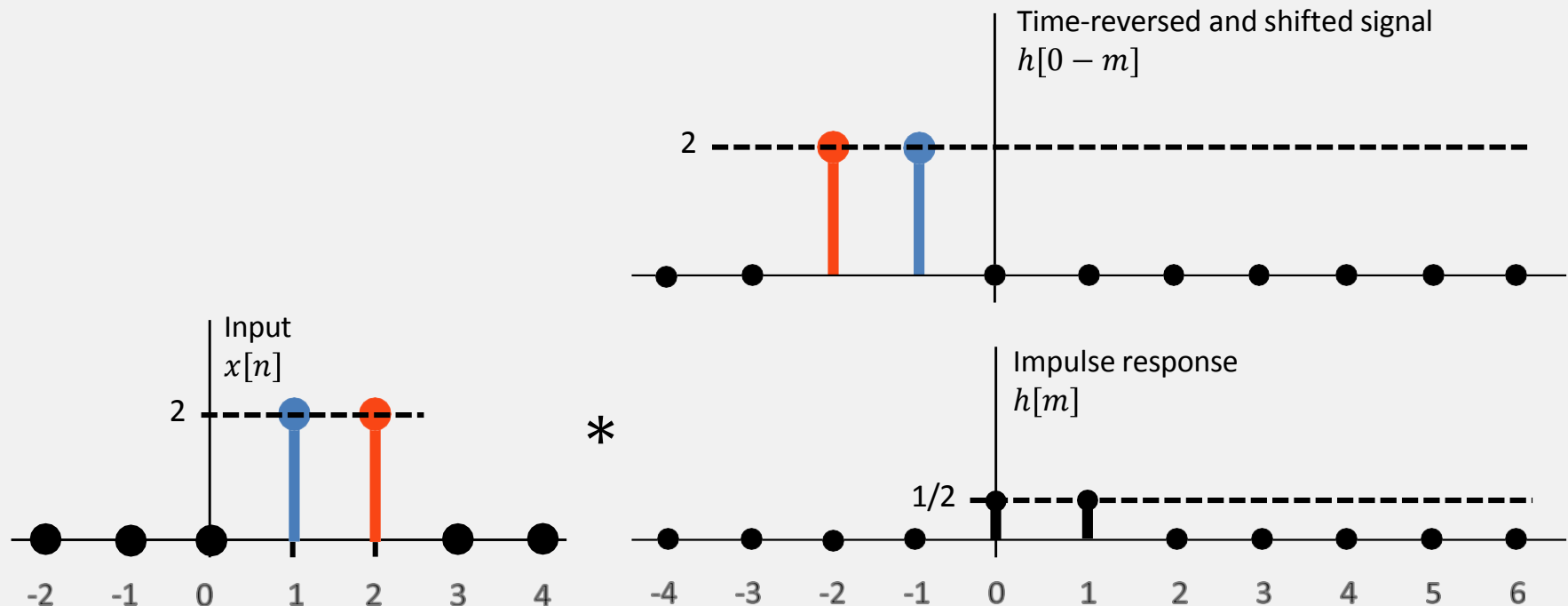
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Step 1: Time-reverse a signal



Convolution

■ Compute the convolution of:

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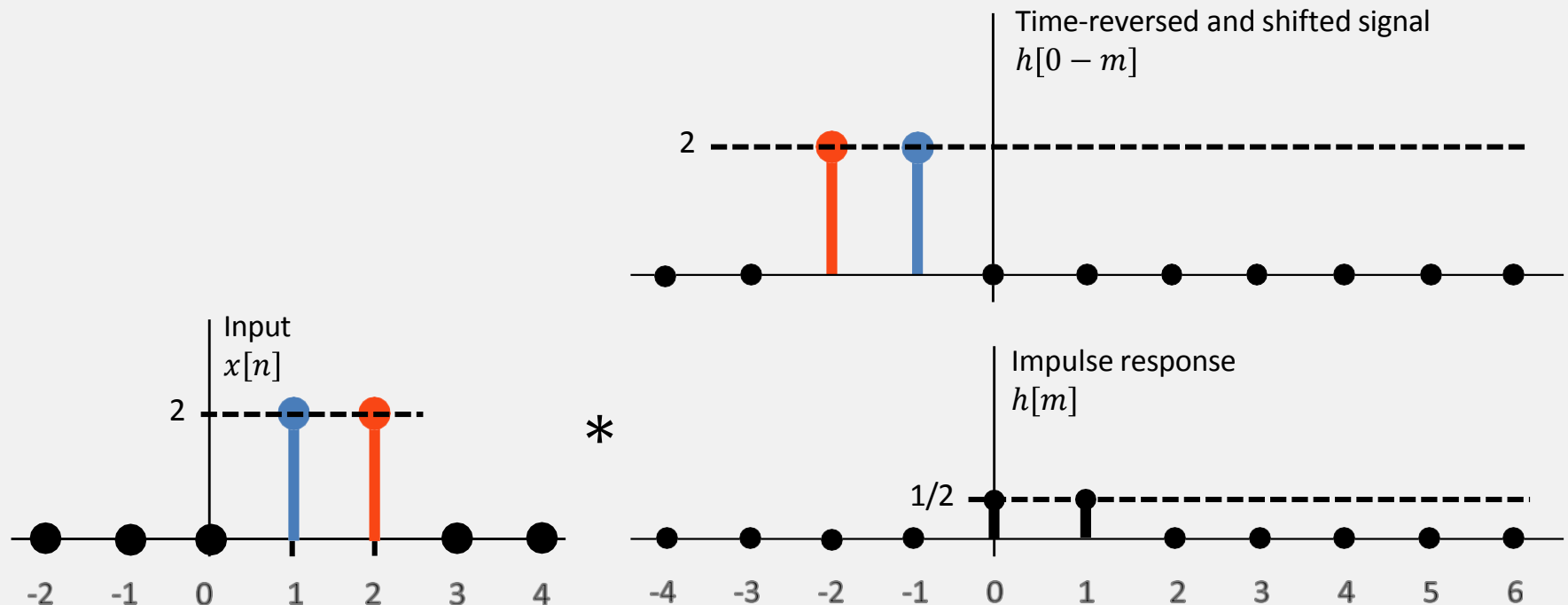
$$h[n] = (1/2)(\delta[n] + \delta[n - 1])$$

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Step 1: Time-reverse a signal

Step 2: Shift that signal by n

Step 3: Multiply the signals & sum the result



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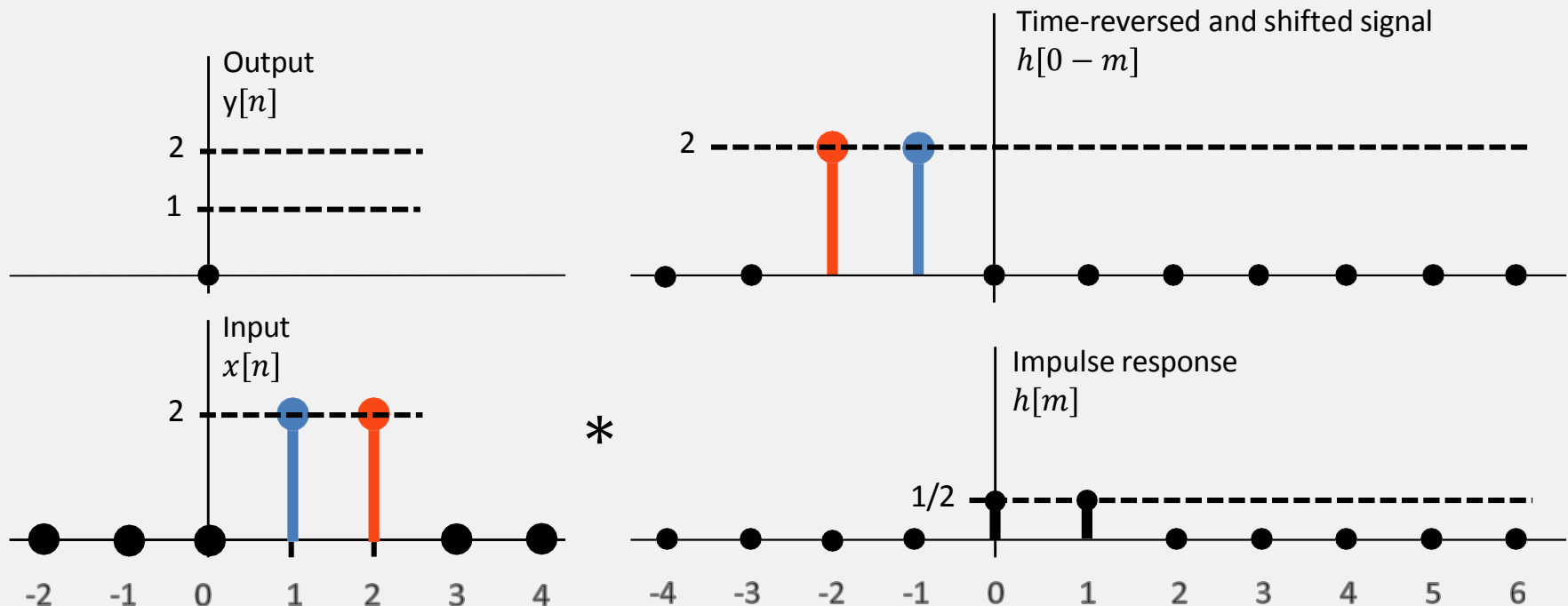
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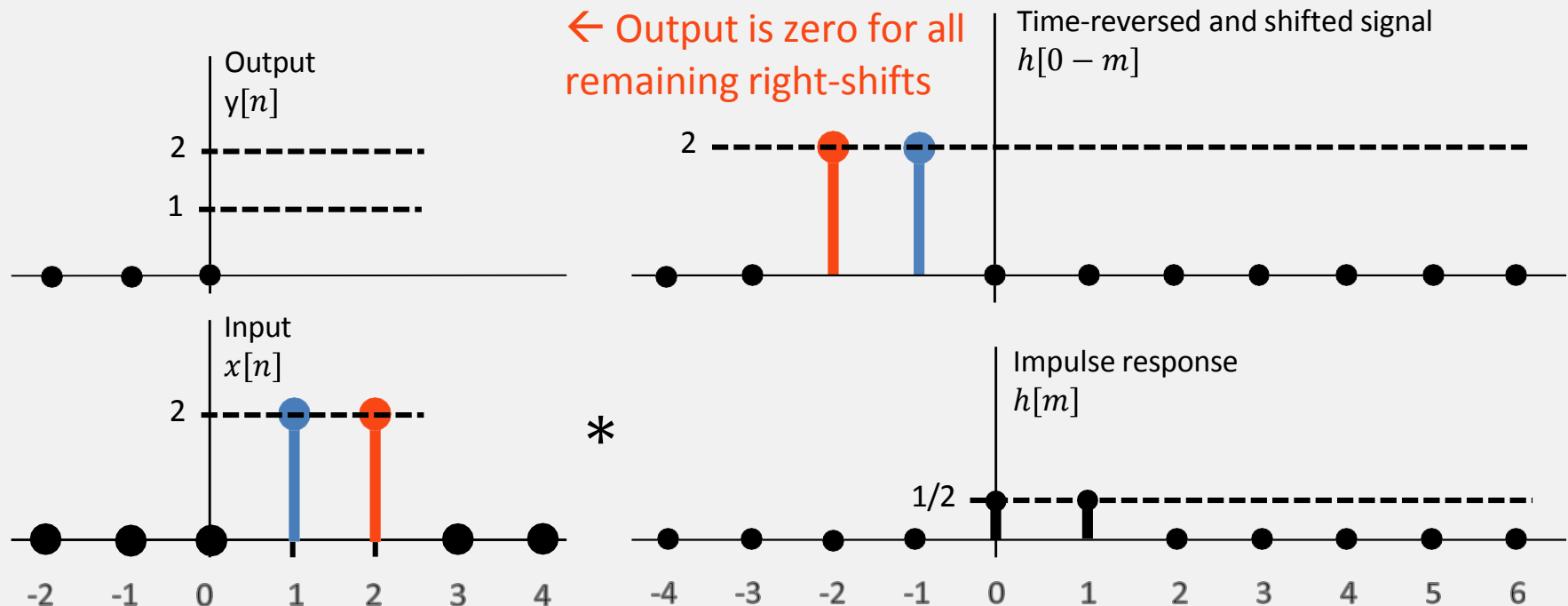
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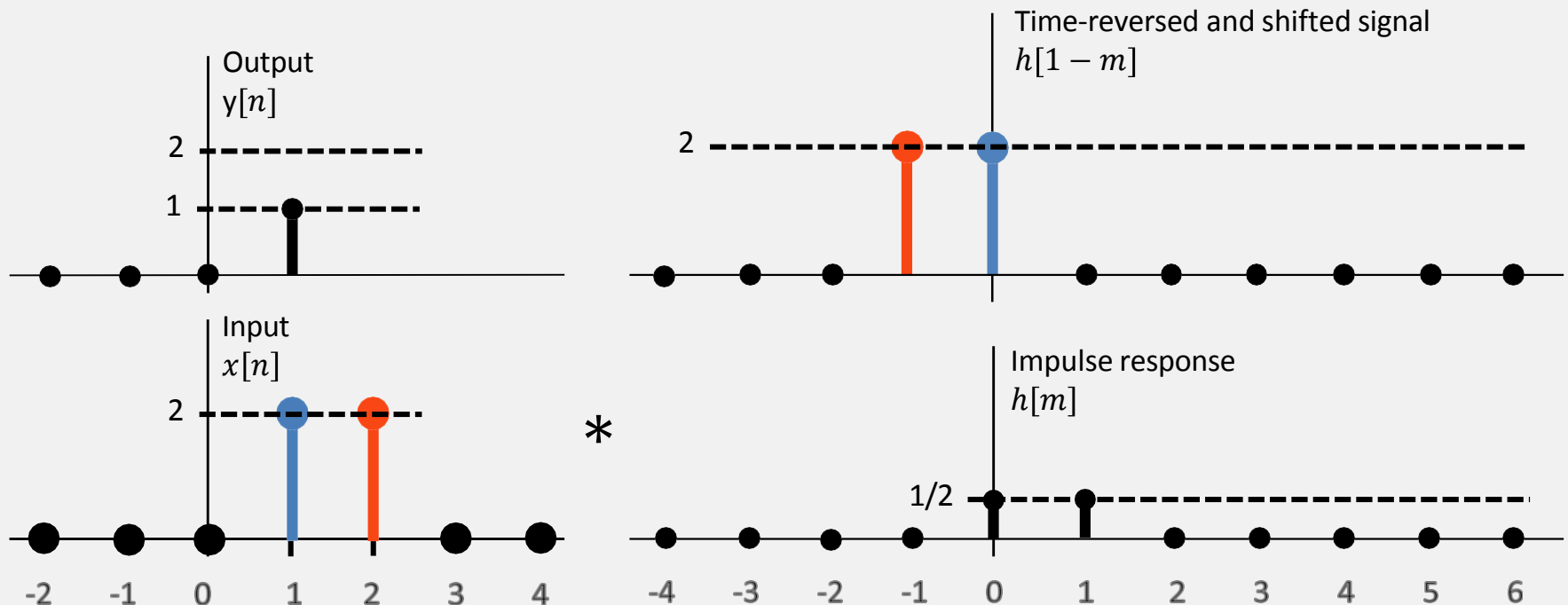
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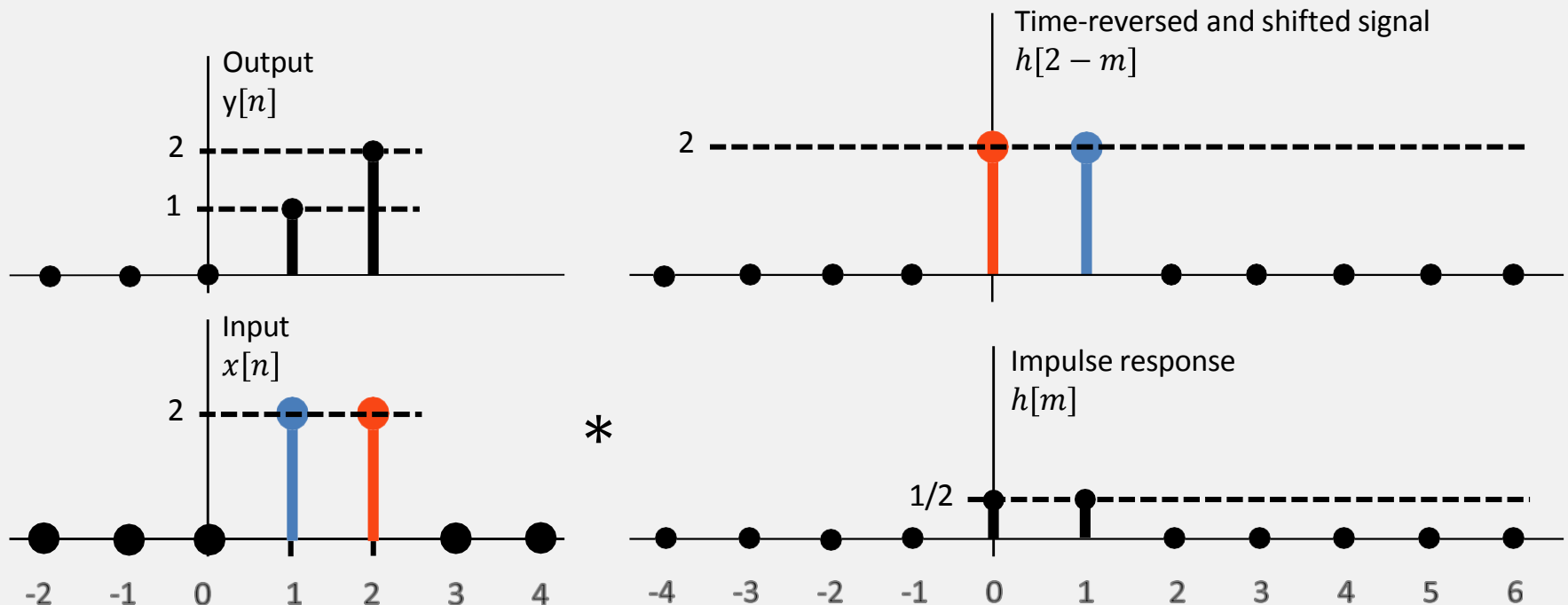
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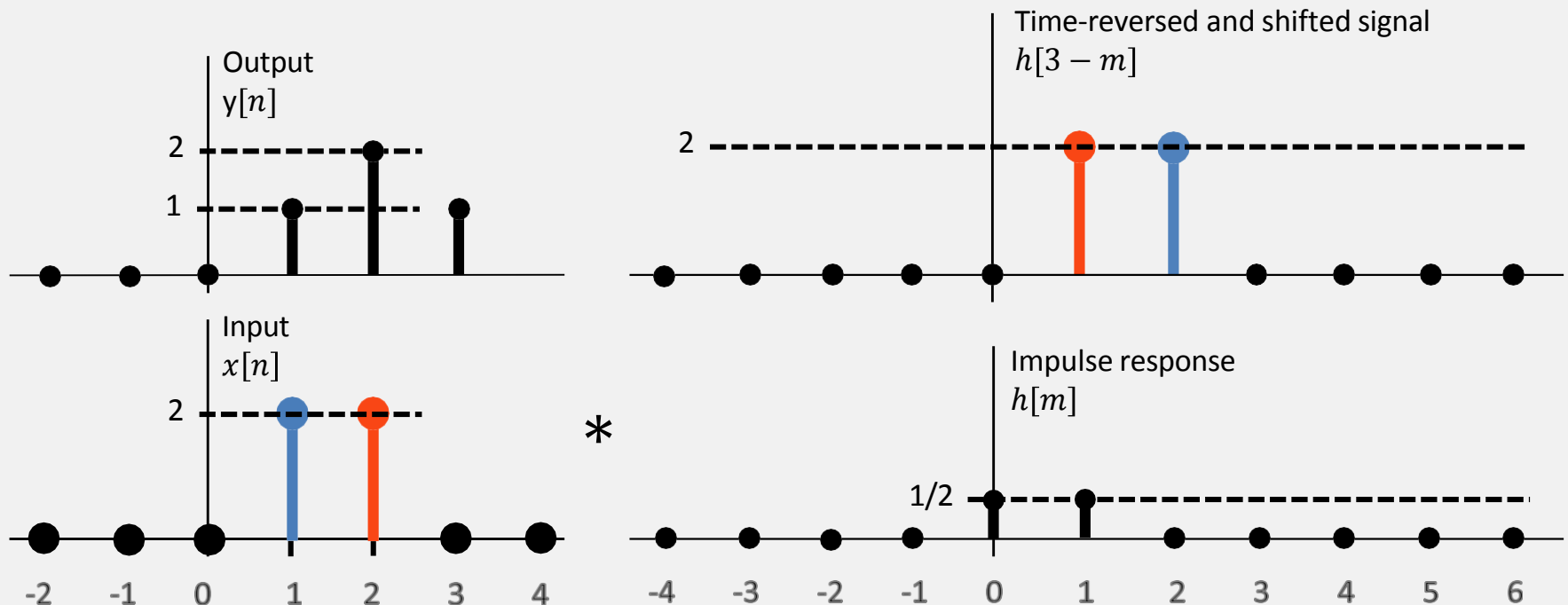
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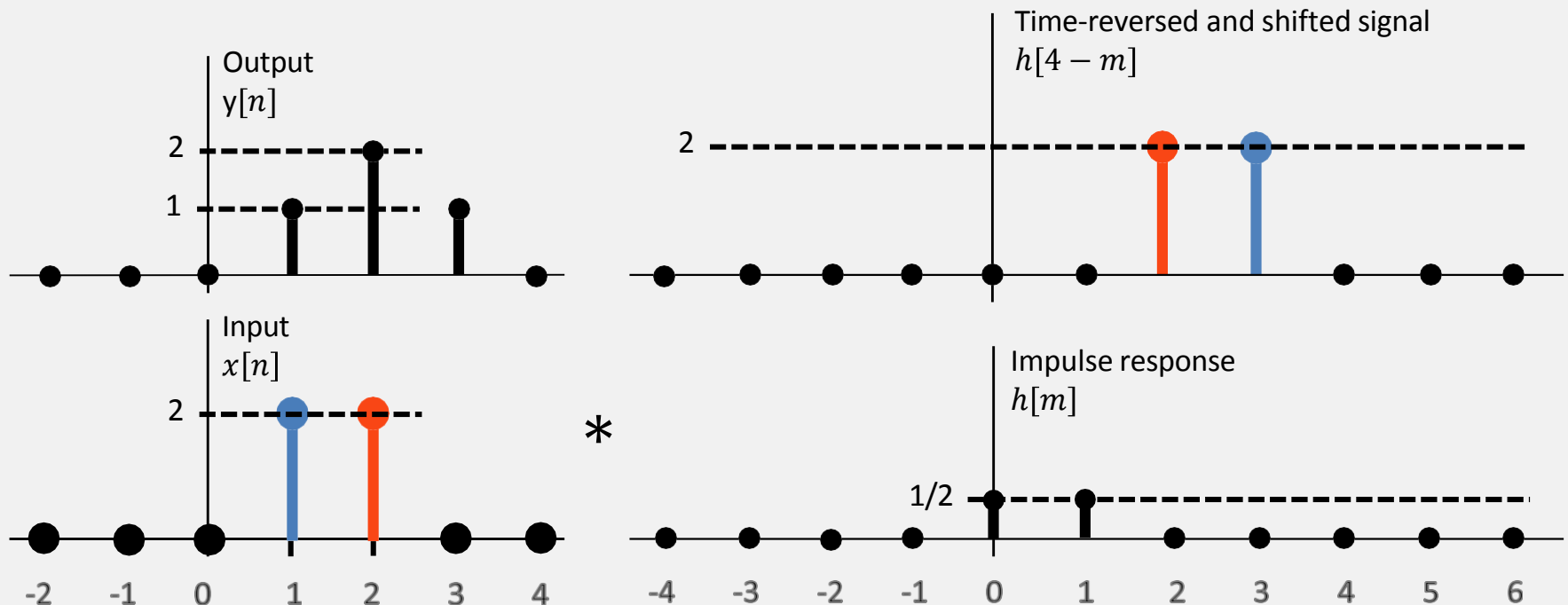
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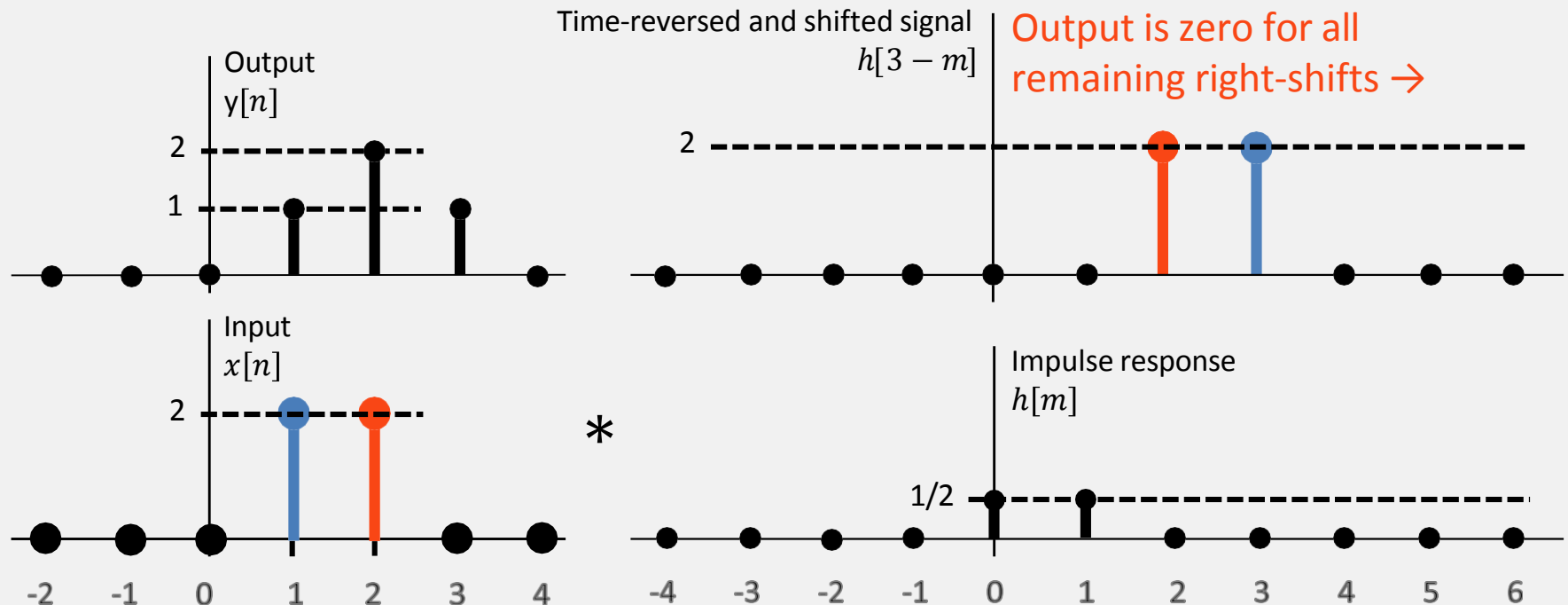
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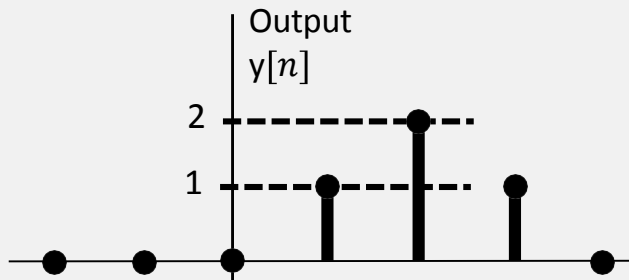
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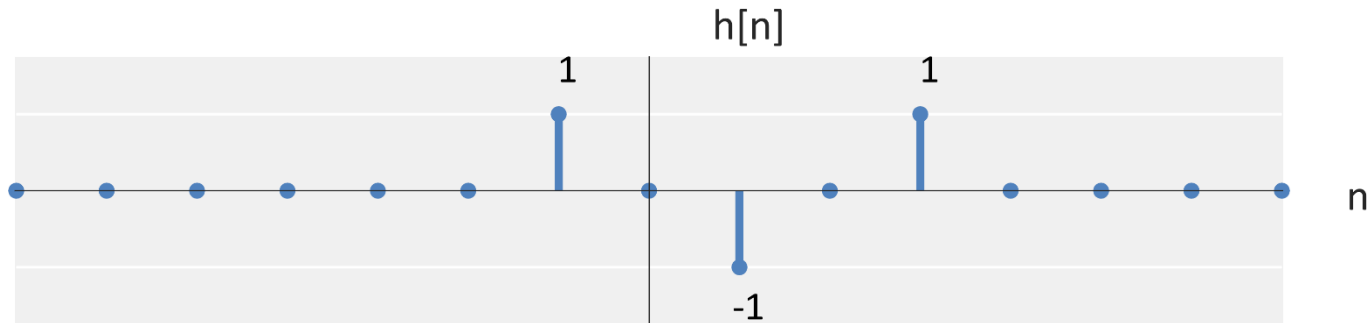
$$y[n] = \delta[n - 1] + 2\delta[n - 2] + \delta[n - 3]$$

Same as our first result!

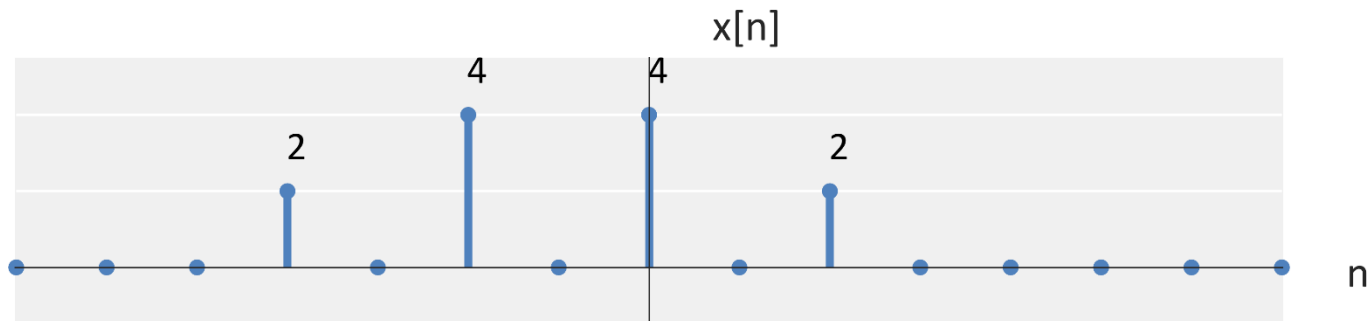
We can compute the system output via convolution!

■ Linear and Time-Invariant (LTI) System

- Consider the system with impulse response $h[n]$



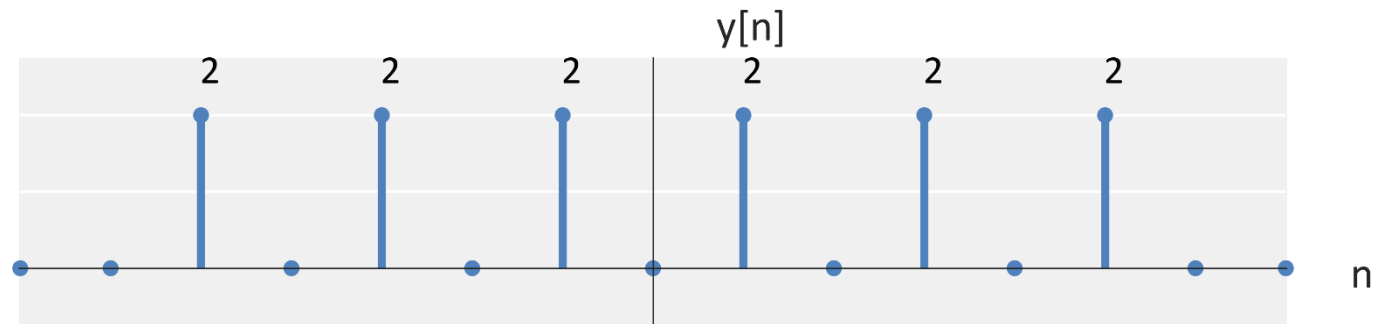
- Compute response of the system to input $x[n]$ below



Convolution

■ Linear and Time-Invariant (LTI) System

■ Solution:



Convolution / Correlation

■ Definition of convolution

$$y[n] = x[n] * h[n] = \sum_{m=-\infty}^{\infty} \text{◇?} x[m] h[n - m]$$

■ Definition of correlation

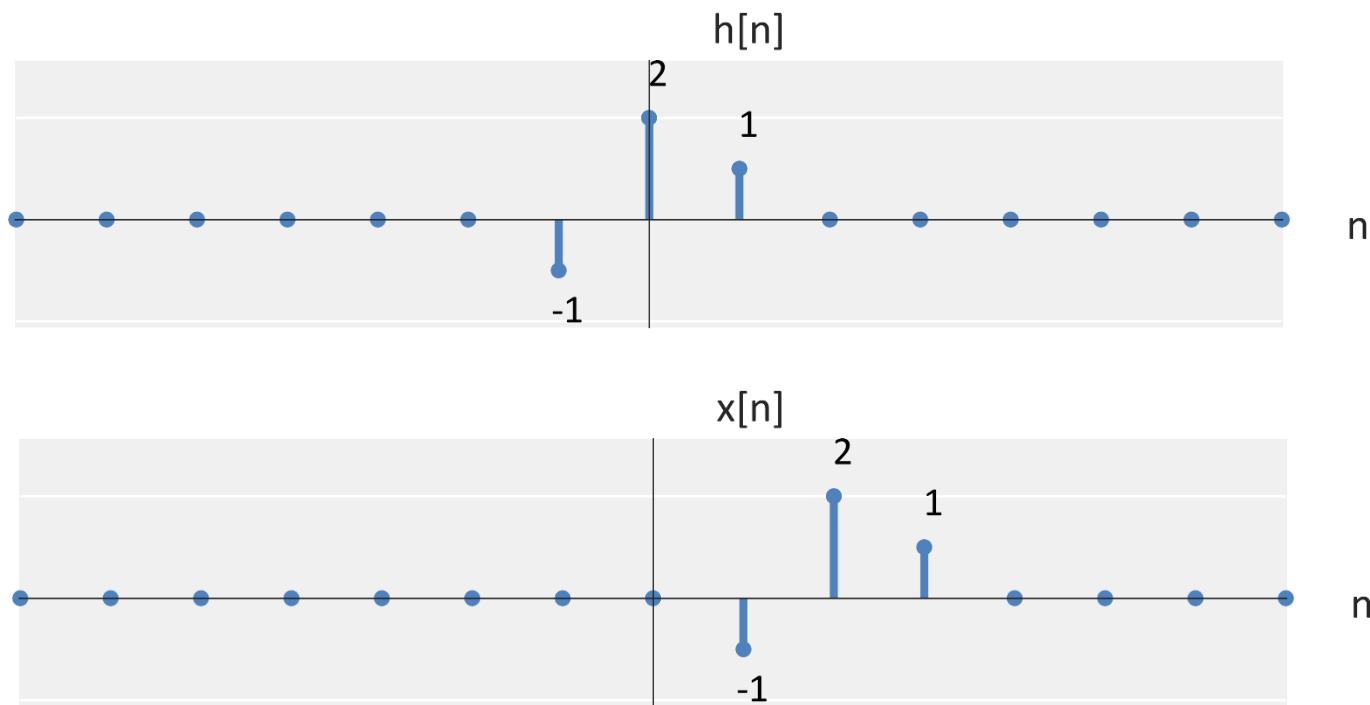
$$y[n] = x[-n] * h[n] = \sum_{m=-\infty}^{\infty} \text{◇?} x[m] h[n + m]$$

Convolution / Correlation

■ Linear and Time-Invariant (LTI) System

- Consider $h[n]$ and $x[n]$ below. Compute their correlation.

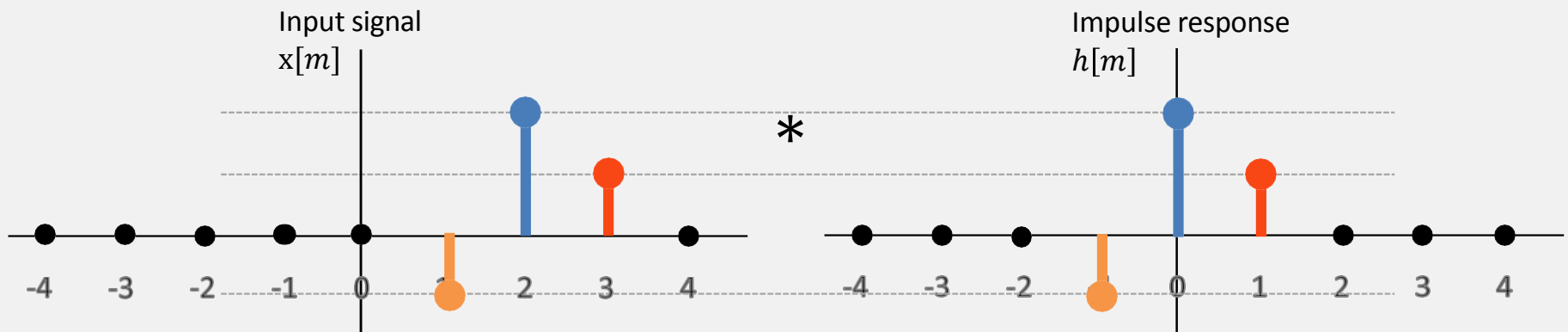
$$y[n] = x[-n] * h[n]$$



Convolution / Correlation

■ Compute the correlation:

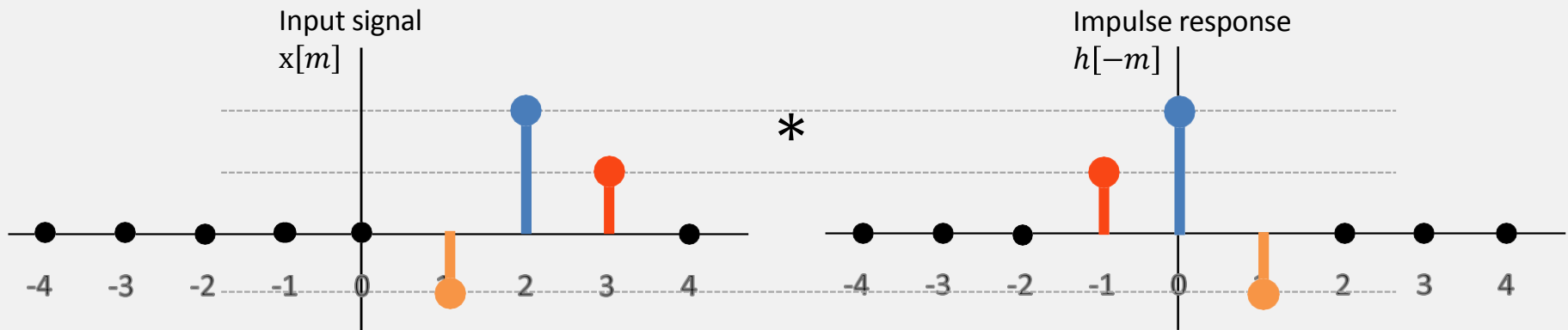
$$y[n] = x[-n] * h[n]$$



Convolution / Correlation

■ Compute the correlation:

$$y[n] = x[-n] * h[n]$$

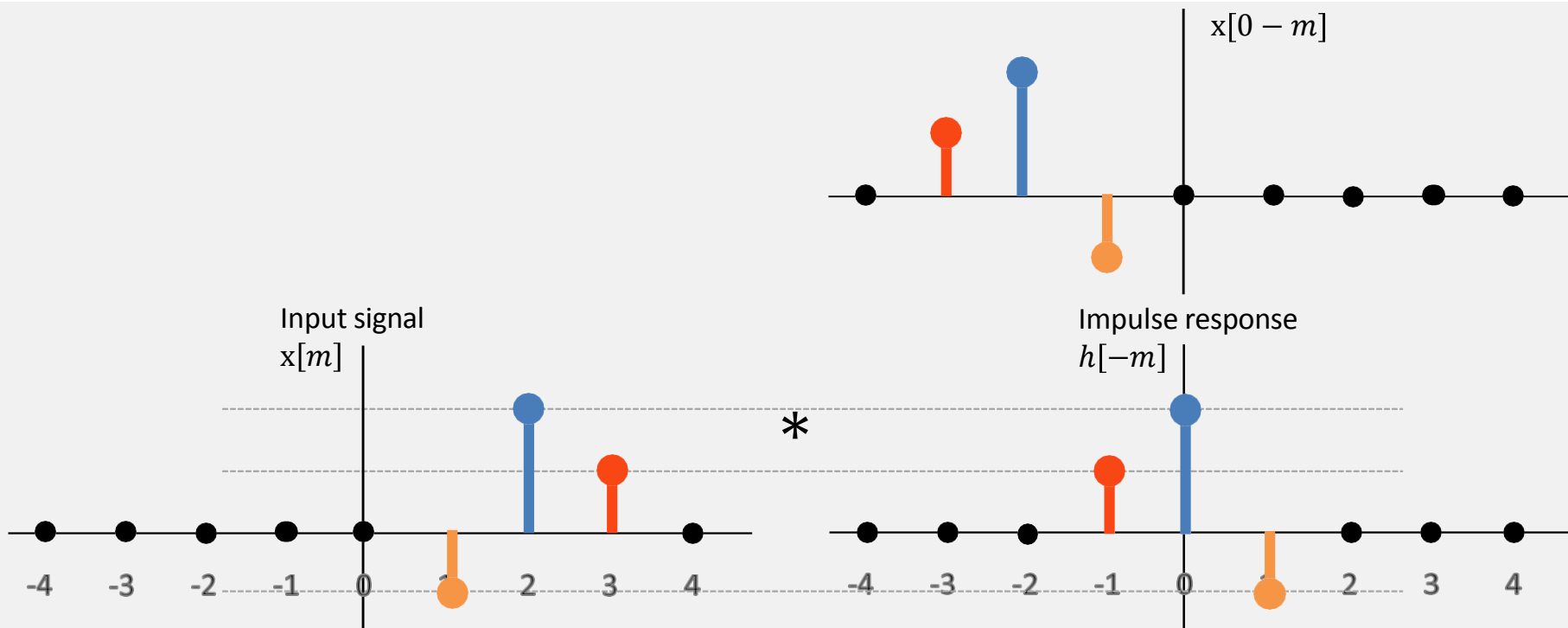


Convolution / Correlation

■ Compute the correlation:

$$y[n] = x[-n] * h[n]$$

Step 1: Time-reverse a signal



Convolution / Correlation

■ Compute the correlation:

$$y[n] = x[-n] * h[n]$$

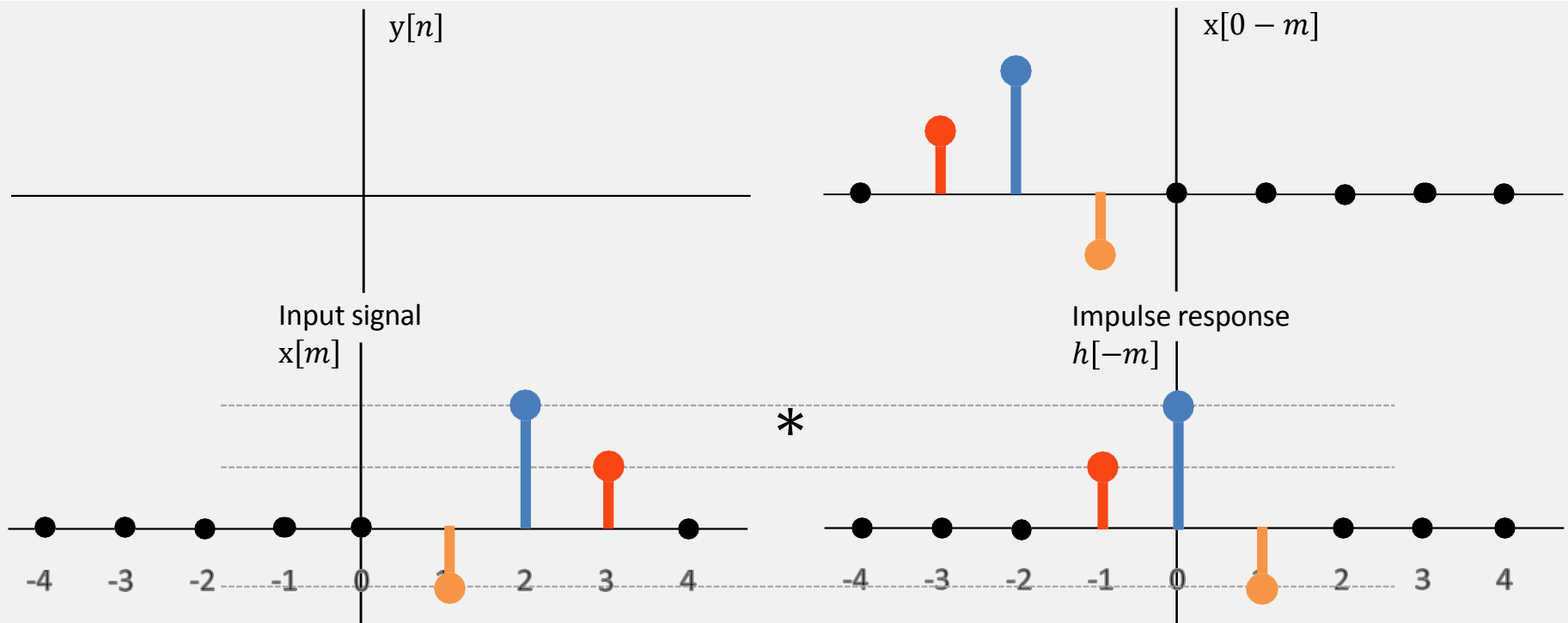
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Step 3: Multiply the signals & sum the result

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Step 5: Repeat for all shifts $-\infty < n < \infty$



Convolution / Correlation

■ Compute the correlation:

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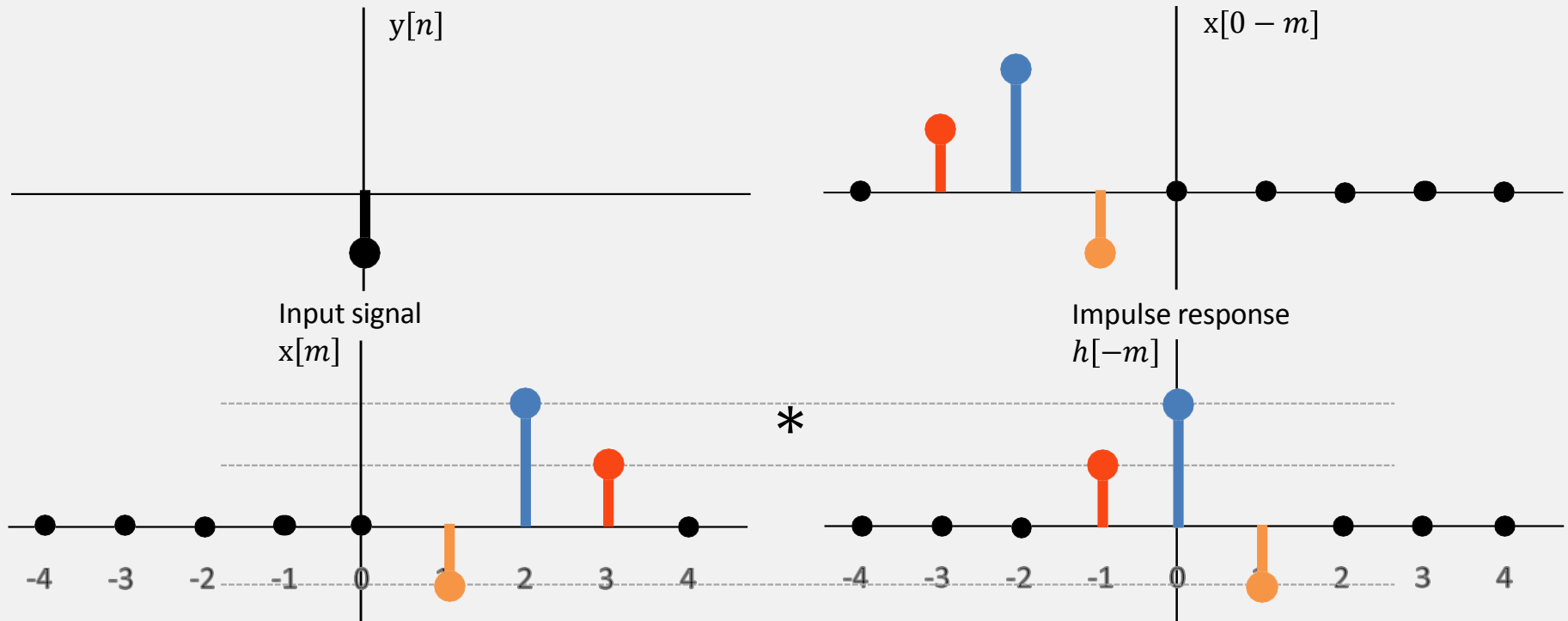
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Convolution / Correlation

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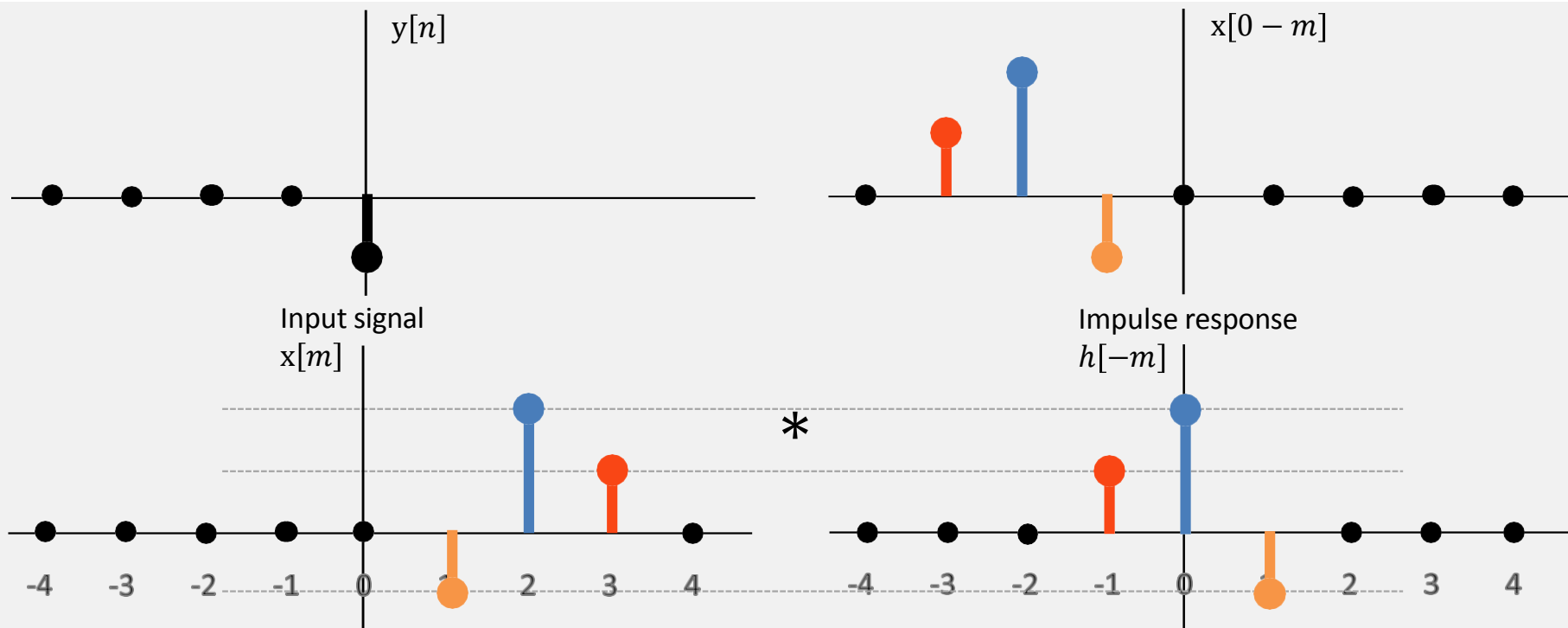
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Convolution / Correlation

■ Compute the correlation:

$$y[n] = x[-n] * h[n]$$

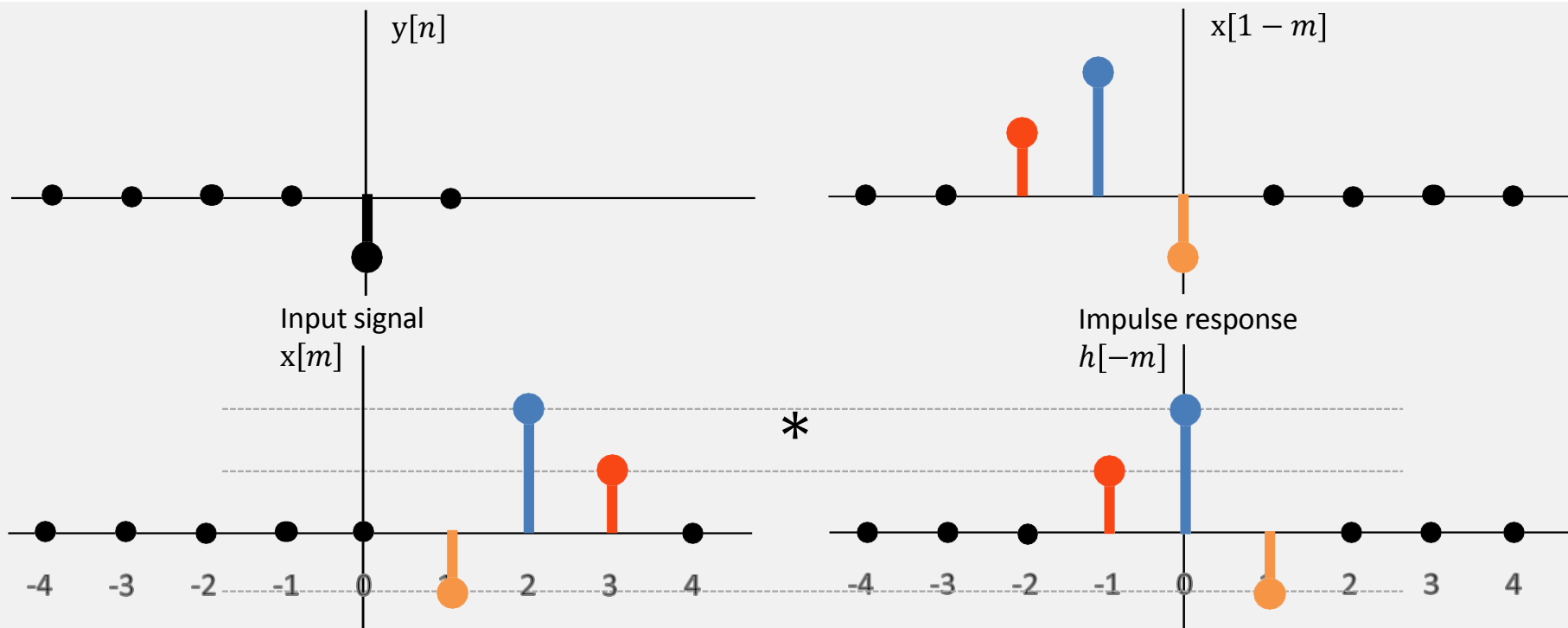
Step 1: Time-reverse a signal

Step 2: Shift that signal by n

Step 3: Multiply the signals & sum the result

Step 4: Assign the sum to $y[n]$ for shift n

Step 5: Repeat for all shifts $-\infty < n < \infty$



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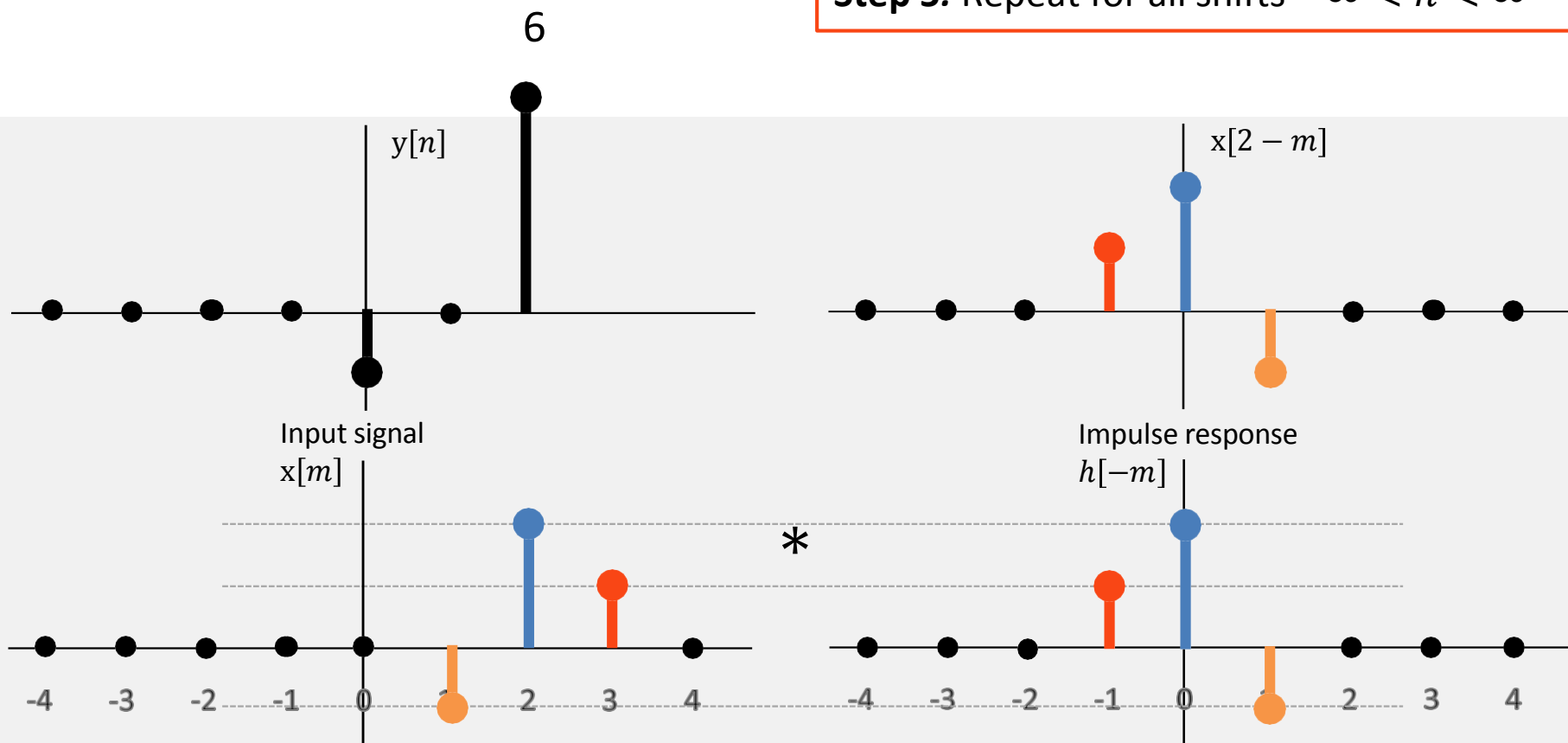
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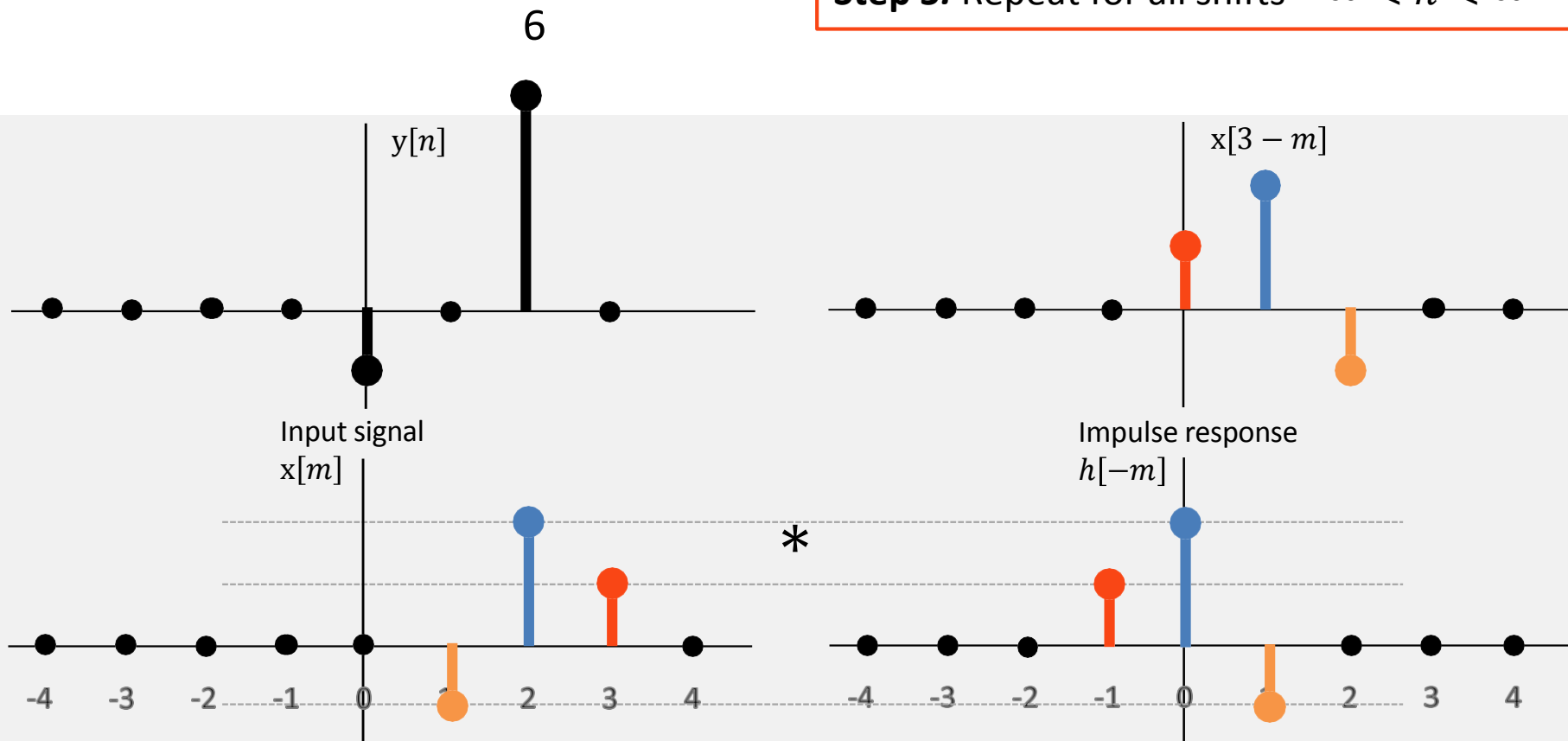
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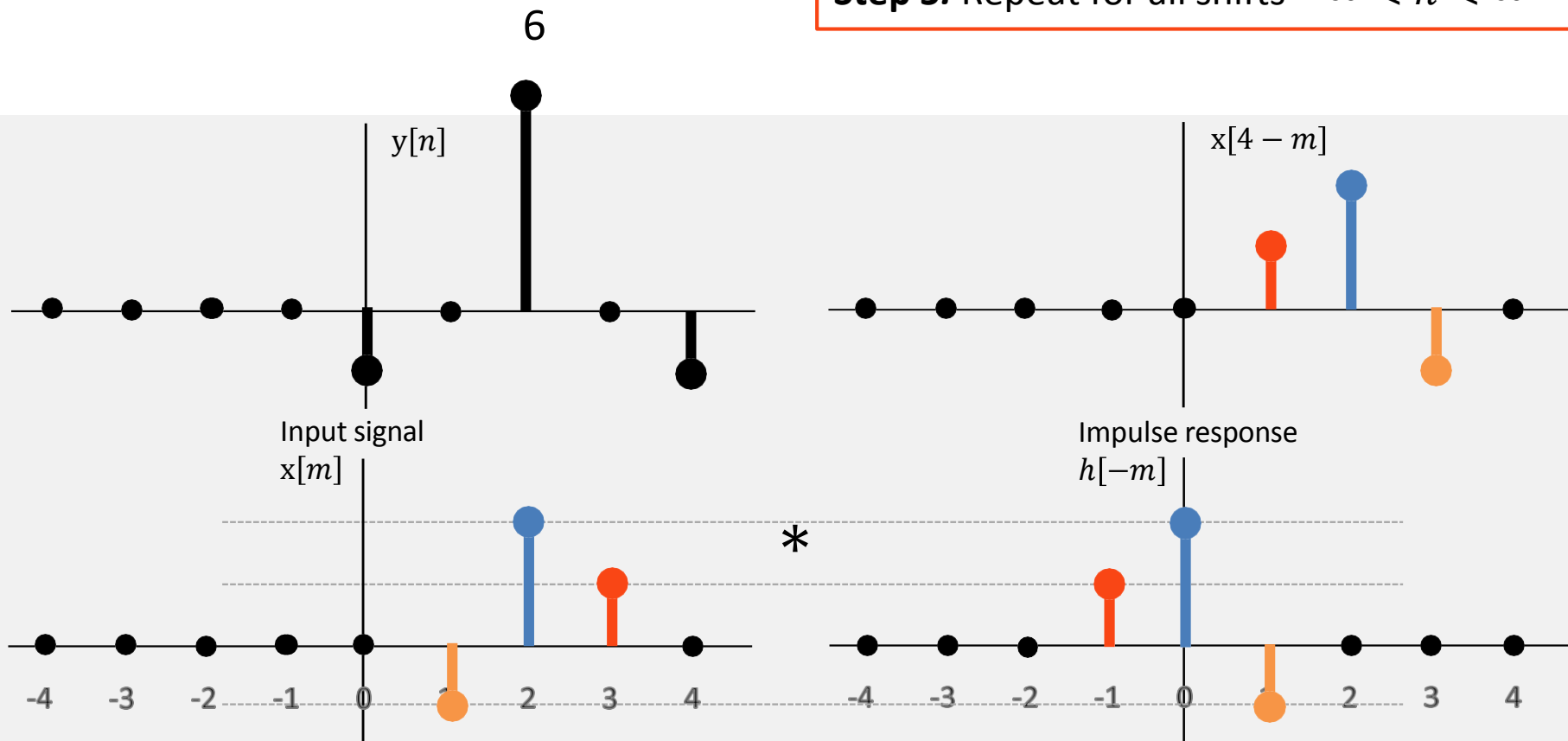
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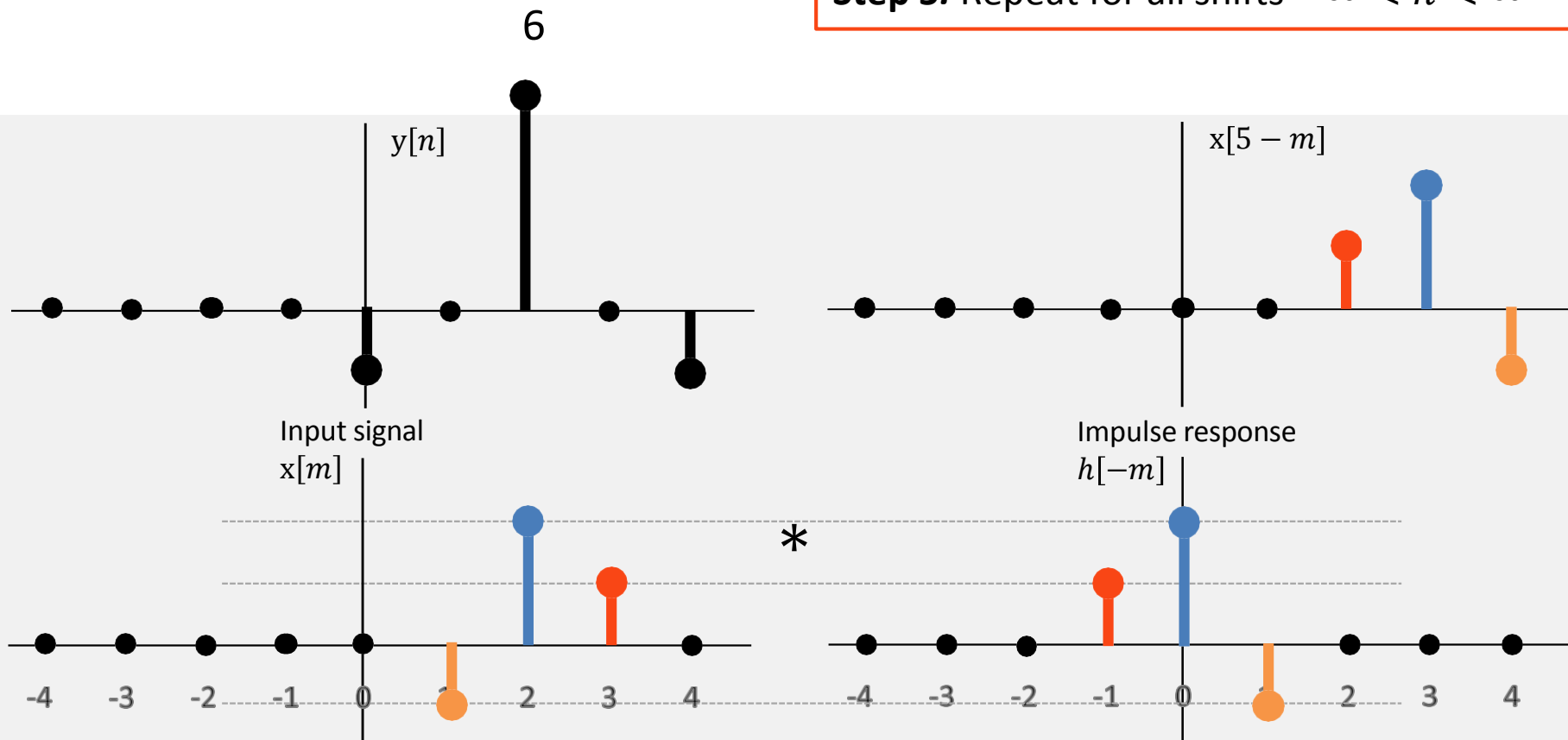
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Convolution / Correlation

■ What does this achieve?

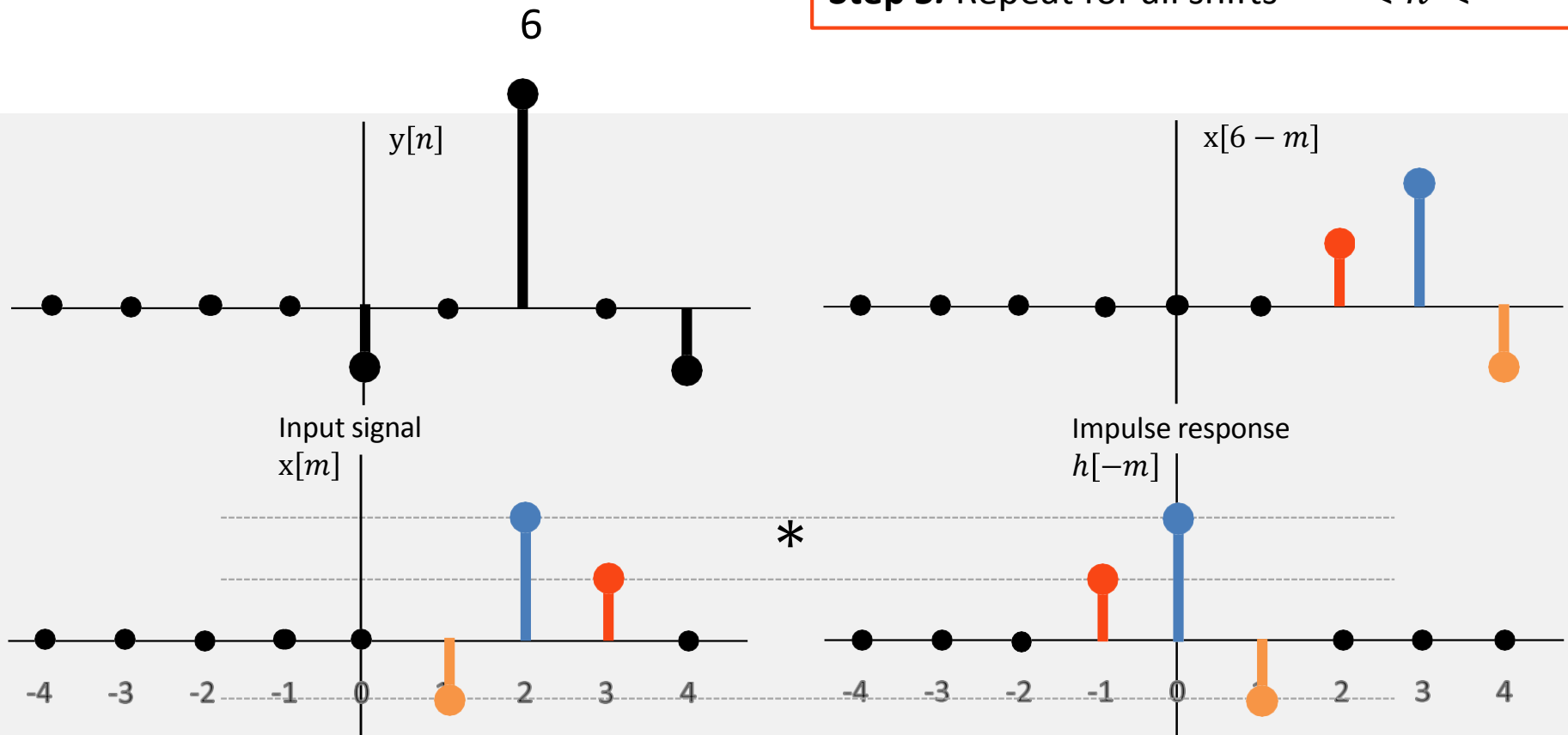
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Thank you